

Plate Tectonics: A Unifying Theory

Unifying Theory

- A unifying theory is one that helps
 - explain a broad range of diverse observations
 - interpret many aspects of a science on a grand scale
 - and relate many seemingly unrelated phenomena
- Plate tectonics is a unifying theory for geology.

Plate Tectonics

■ Plate tectonics helps to explain

- ❑ earthquakes
- ❑ volcanic eruptions
- ❑ formation of mountains
- ❑ location of continents
- ❑ location of ocean basins

■ Tectonic interactions affect

- ❑ atmospheric and oceanic circulation and climate
- ❑ geographic distribution,
- ❑ evolution and extinction of organisms
- ❑ distribution and formation of resources

Early Ideas about Continental Drift

■ Edward Suess

- Austrian, late 1800s
- noted similarities between
- the Late Paleozoic plant fossils
 - *Glossopteris* flora
- and evidence for glaciation
- in rock sequences of
 - India
 - Australia
 - South Africa
 - South America



- He proposed the name *Gondwanaland* (or **Gondwana**)
 - for a supercontinent
 - composed of these continents

Early Ideas about Continental Drift

■ Frank Taylor (American, 1910)

- presented a hypothesis of continental drift with these features:
 - lateral movement of continents formed mountain ranges
 - a continent broke apart at the Mid-Atlantic Ridge to form the Atlantic Ocean
 - supposedly, tidal forces pulled formerly polar continents toward the equator,
 - when Earth captured the Moon about 100 million years ago

Alfred Wegener and the Continental Drift Hypothesis

- German meteorologist
- Credited with hypothesis of continental drift-1912 in a scientific presentation – published a book in 1915.



Alfred Wegener and the Continental Drift Hypothesis

- He proposed that all landmasses
 - were originally united into a supercontinent
 - he named Pangaea from the Greek meaning “all land”
- He presented a series of maps
 - showing the breakup of Pangaea
- He amassed a tremendous amount of geologic, paleontologic, and climatologic evidence

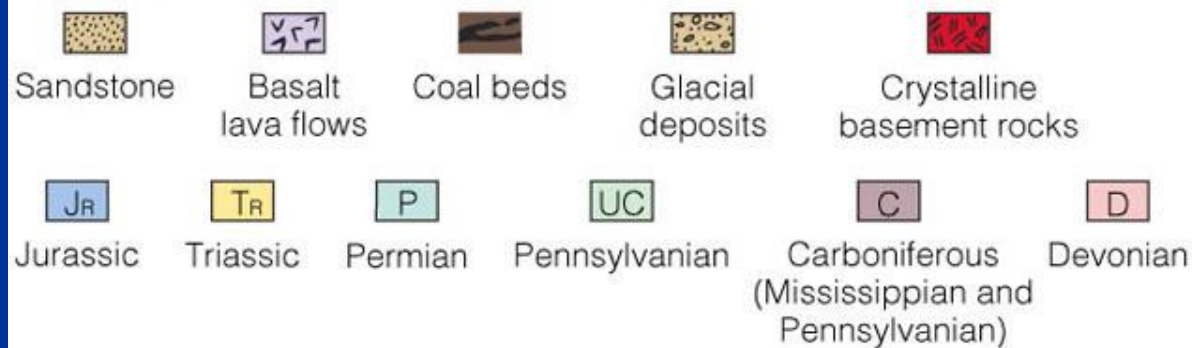
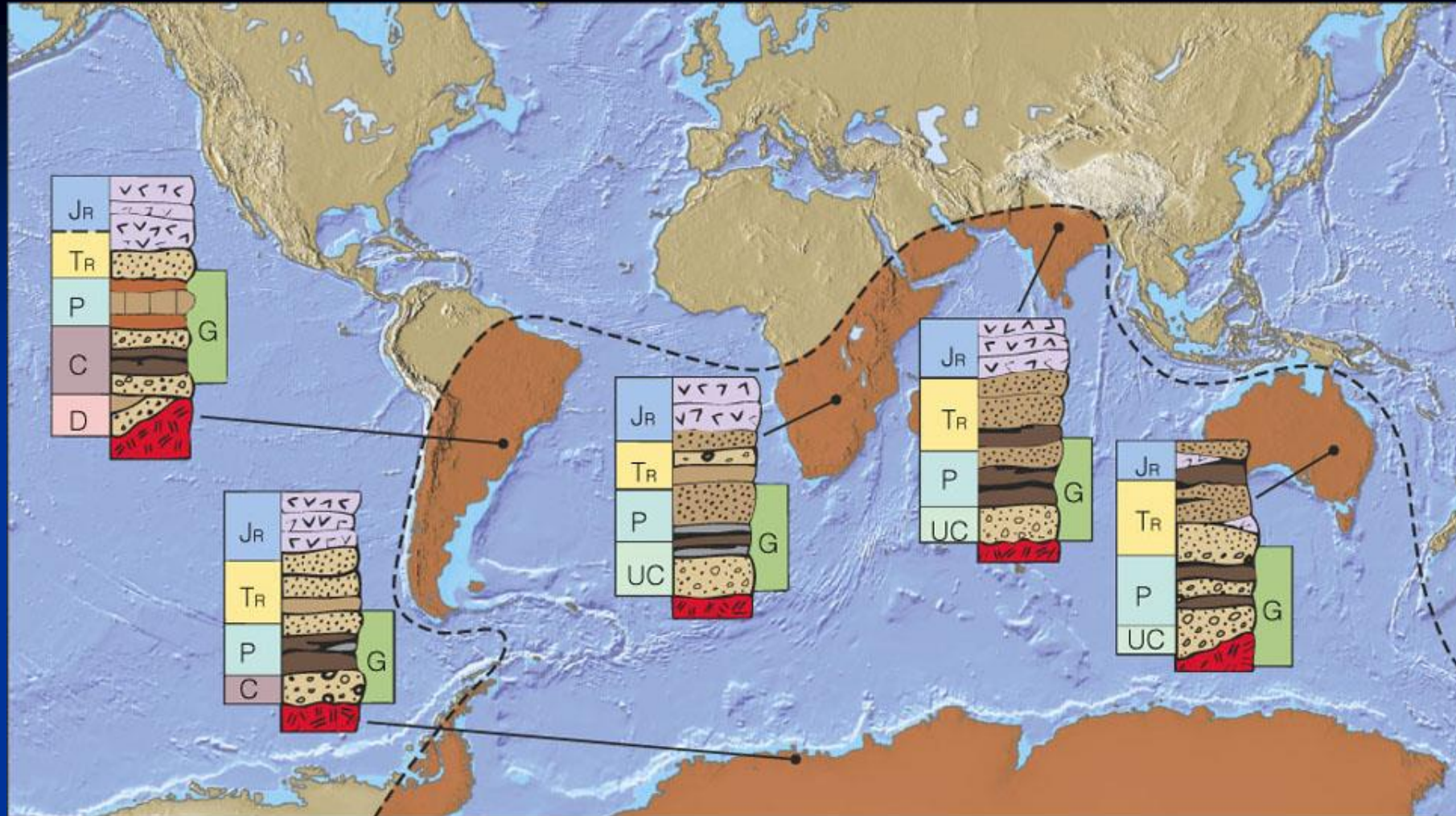
Wegener's Evidence

- Shorelines of continents fit together
 - matching marine, nonmarine
 - and glacial rock sequences
 - from Pennsylvanian to Jurassic age
 - for all five Gondwana continents
 - including Antarctica
- Mountain ranges and glacial deposits
 - match up when continents are united
 - into a single landmass

Jigsaw-Puzzle Fit of Continents

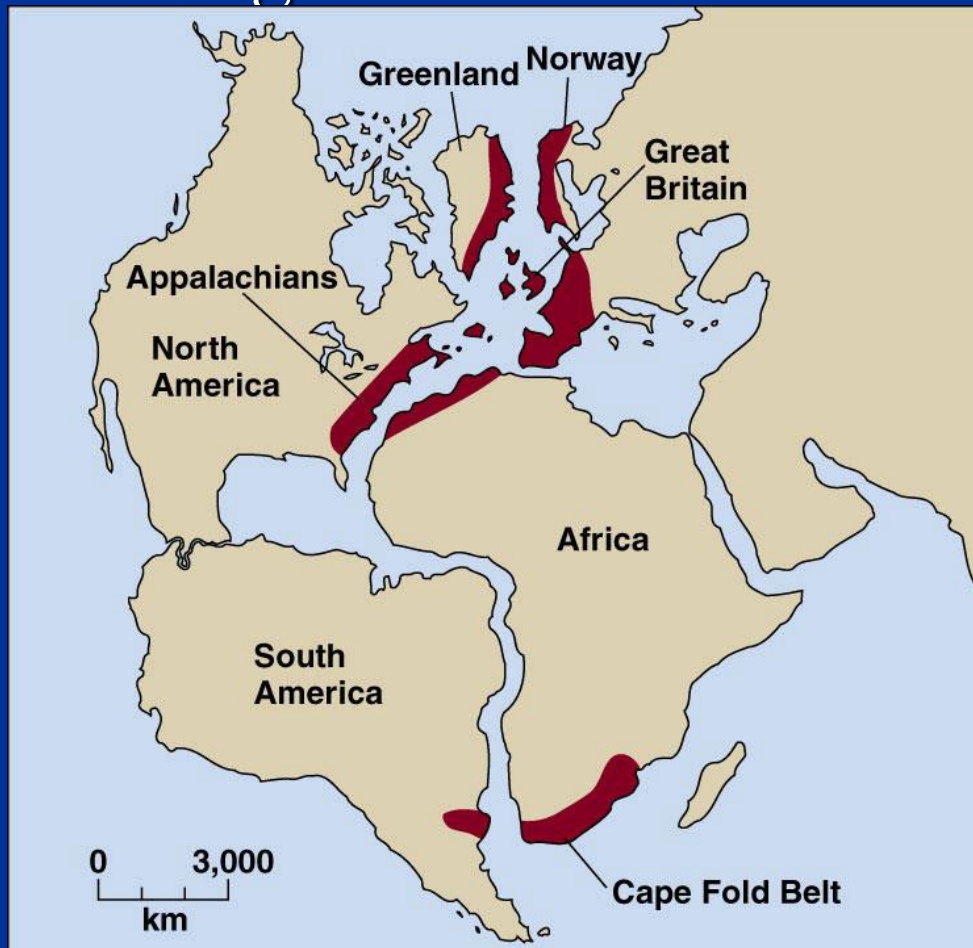
■ Continental Fit





Jigsaw-Puzzle Fit of Continents

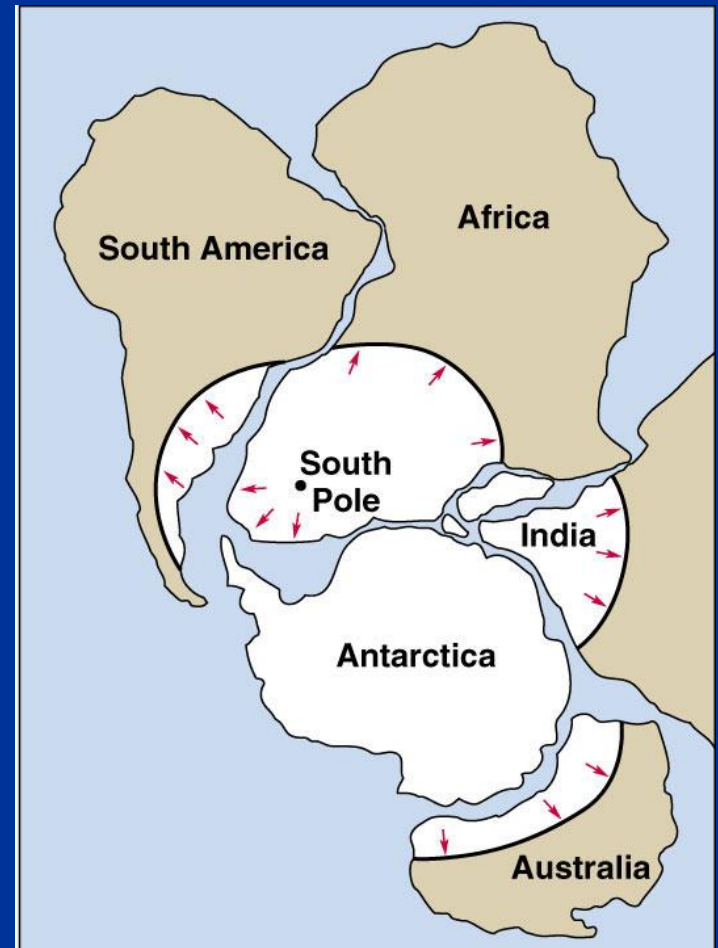
■ Matching mountain ranges



(a)

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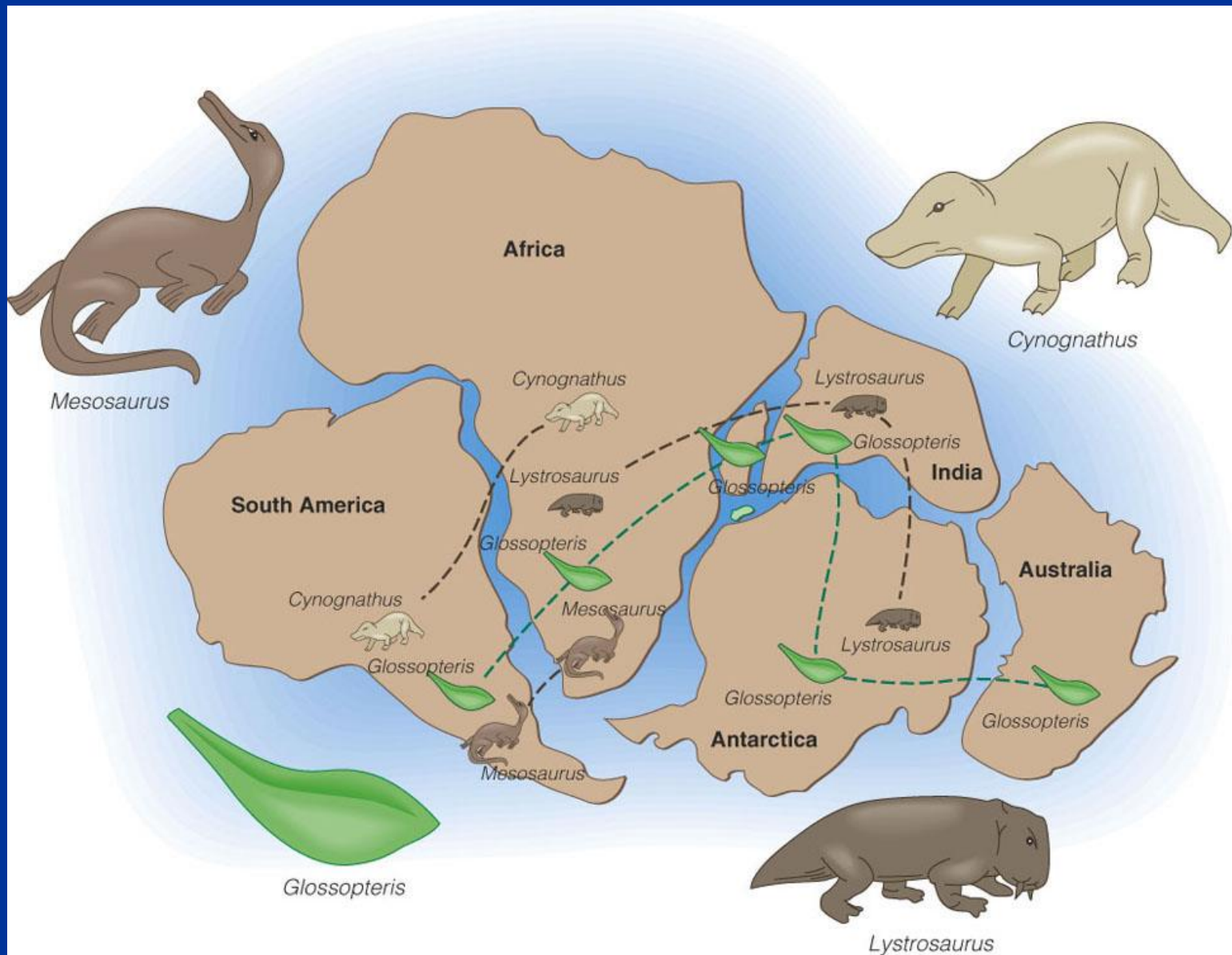
■ Matching glacial evidence



(b)

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Matching Fossils

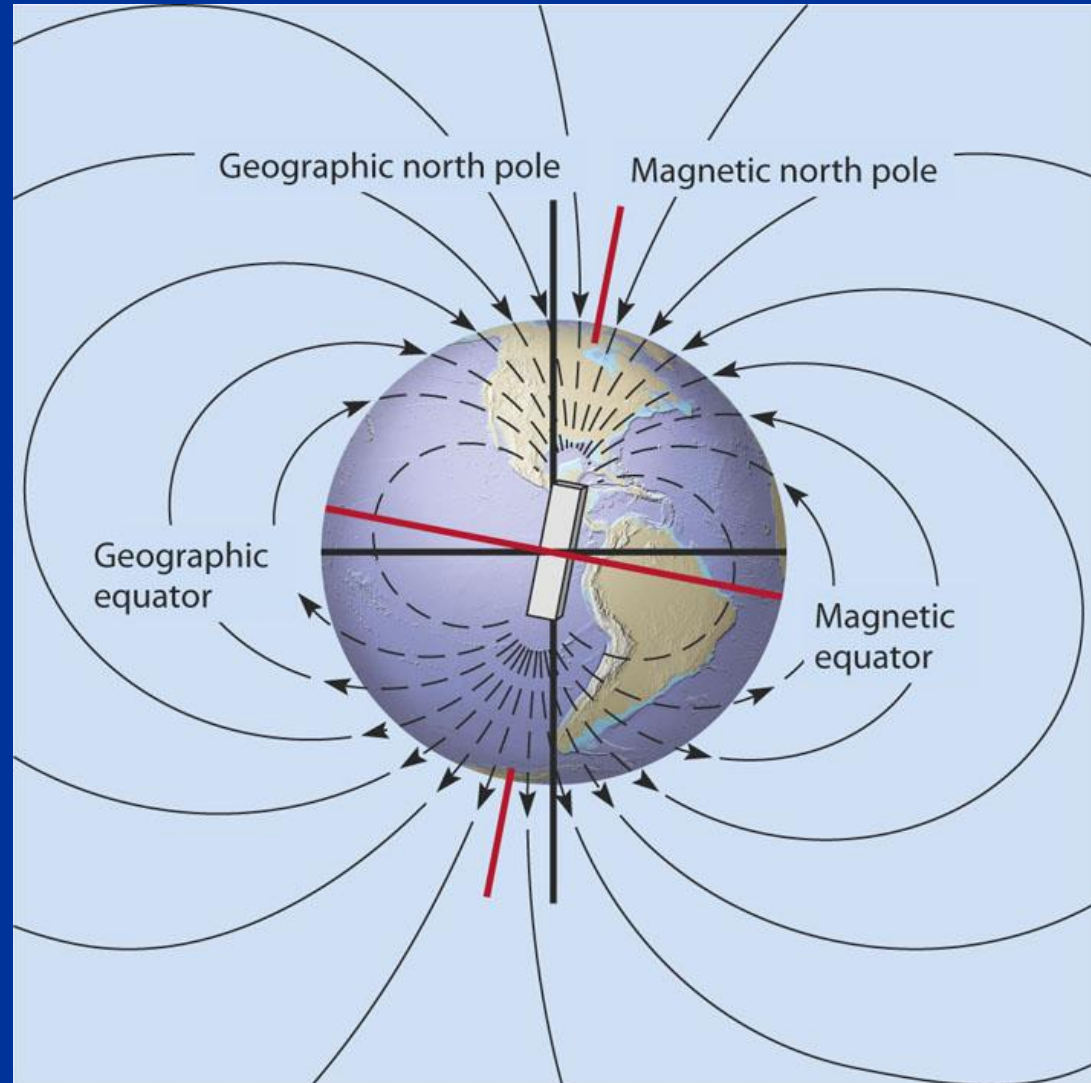


The Perceived Problem with Continental Drift

- Most geologists did not accept the idea of moving continents
 - There was no suitable mechanism to explain
 - how continents could move over Earth's surface
- Interest in continental drift only revived when
 - new evidence from studies of Earth's magnetic field
 - and oceanographic research
 - showed that the ocean basins were geologically young features

Earth's Magnetic Field

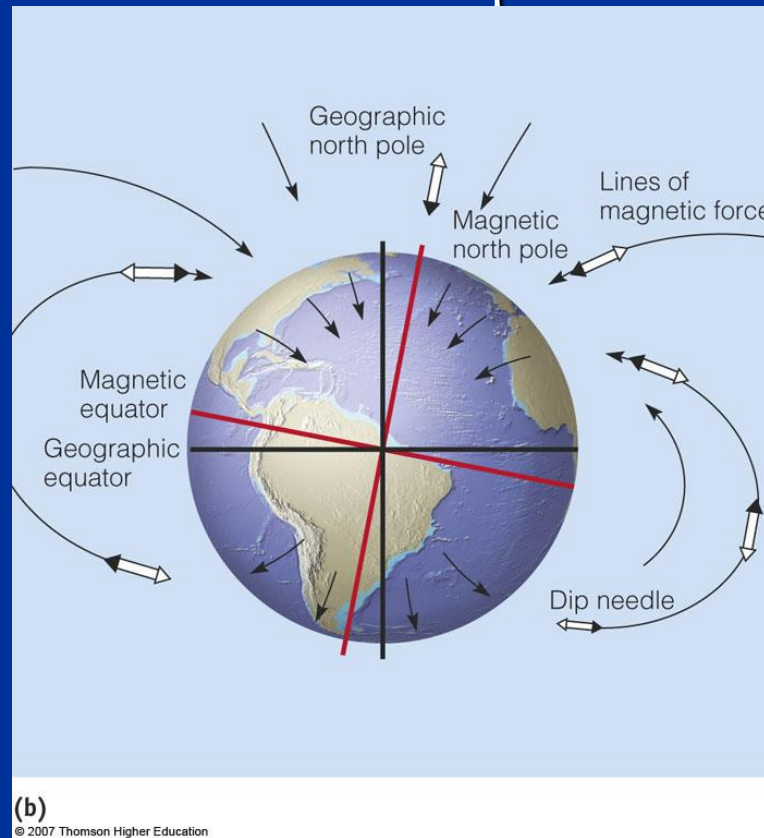
- Earth as a giant dipole magnet
 - magnetic poles essentially coincide
 - with the geographic poles
 - and may result from different rotation speeds
 - of outer core and mantle



(a)

Magnetic Field Varies

- Strength and orientation of the magnetic field varies
 - weak and horizontal at the equator
 - strong and vertical at the poles

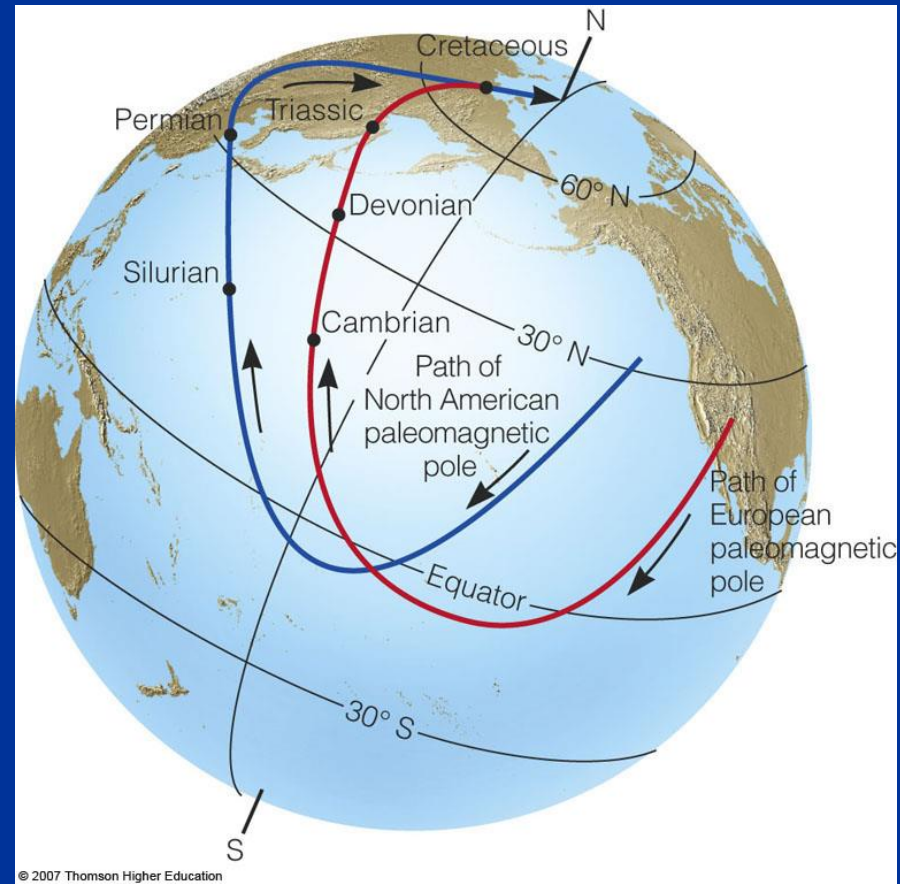


Paleomagnetism

- Paleomagnetism is
 - a remanent magnetism
 - in ancient rocks
 - recording the direction
 - and the strength of Earth's magnetic field
 - at the time of the rock's formation
- When magma cools
 - below the Curie point temperature
 - magnetic iron-bearing minerals align
 - with Earth's magnetic field

Polar Wandering

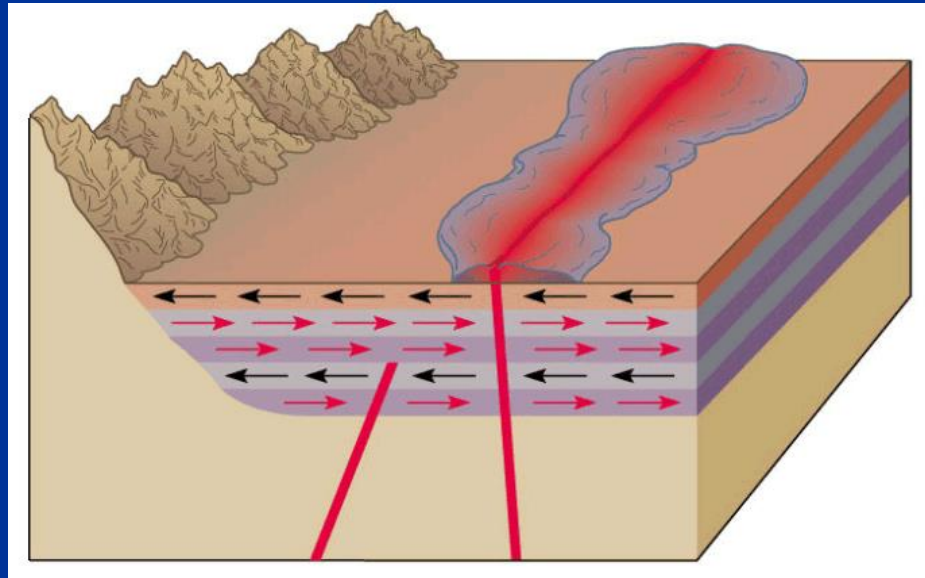
- In 1950s, research revealed
 - that paleomagnetism of ancient rocks showed
 - orientations different from the present magnetic field
- Magnetic poles apparently moved.
 - The apparent movement was called *polar wandering*.
 - Different continents had different paths.
- The best explanation
 - is stationary poles
 - and moving continents



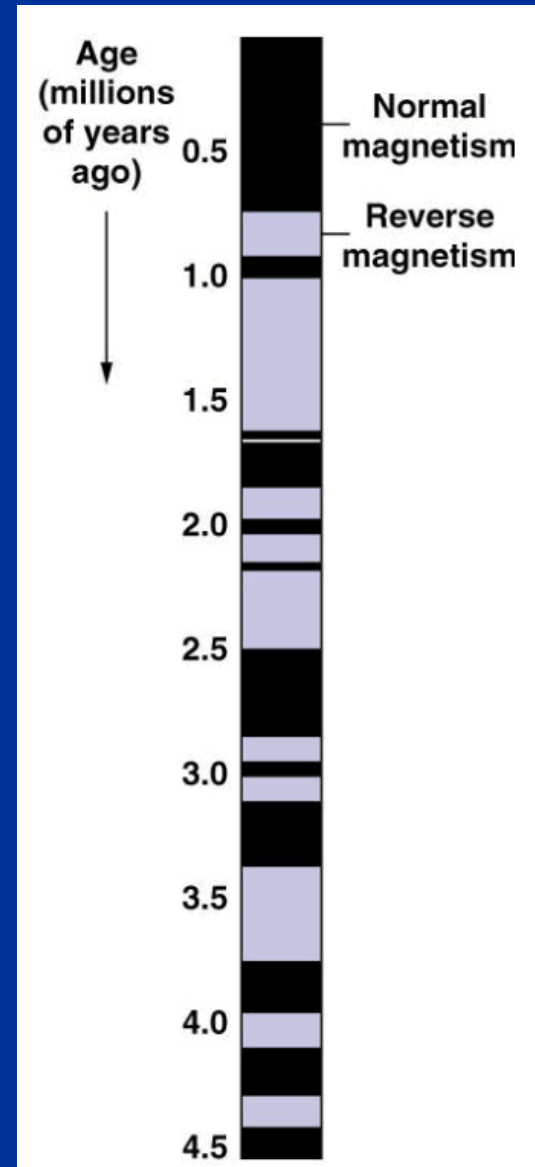
Magnetic Reversals

- Earth's present magnetic field is called **normal**,
 - with magnetic north near the north geographic pole
 - and magnetic south near the south geographic pole
- At various times in the past,
 - Earth's magnetic field has completely **reversed**,
 - with magnetic south near the north geographic pole
 - and magnetic north near the south geographic pole
- This is referred to as a **magnetic reversal**

Magnetic Reversals



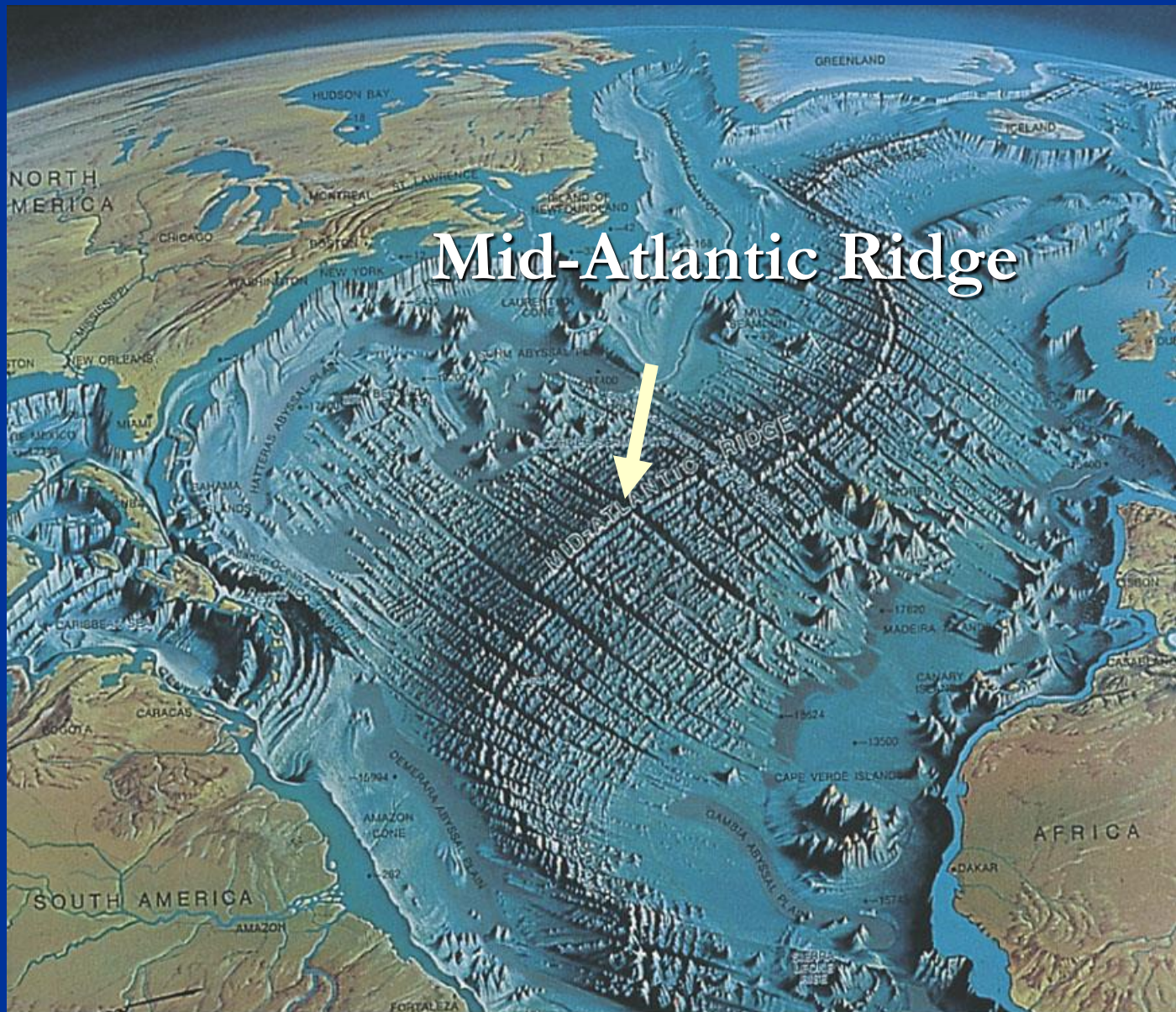
- Measuring paleomagnetism and dating continental lava flows led to
 - the realization that magnetic reversals existed
 - the establishment of a magnetic reversal time scale



Mapping Ocean Basins

- Ocean mapping revealed
 - a ridge system
 - more than 65,000 km long,
 - the most extensive mountain range in the world
- The Mid-Atlantic Ridge
 - is the best known part of the system
 - and divides the Atlantic Ocean basin
 - in two nearly equal parts

Atlantic Ocean Basin

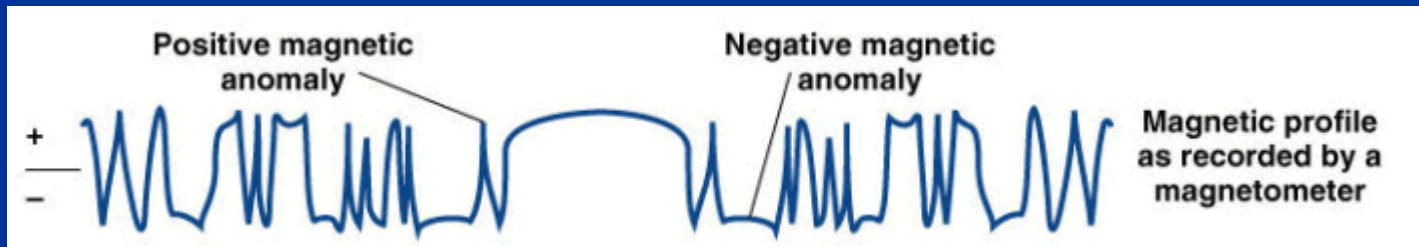


Seafloor Spreading

- **Harry Hess**, in 1962, proposed the theory of seafloor spreading:
 - Continents and oceanic crust move together
 - Seafloor separates at oceanic ridges
 - where new crust forms from upwelling and cooling magma, and
 - the new crust moves laterally away from the ridge
 - The mechanism that drives seafloor spreading was thermal convection cells in the mantle
 - hot magma rises from mantle to form new crust
 - cold crust subducts into the mantle at oceanic trenches, where it is heated and recycled

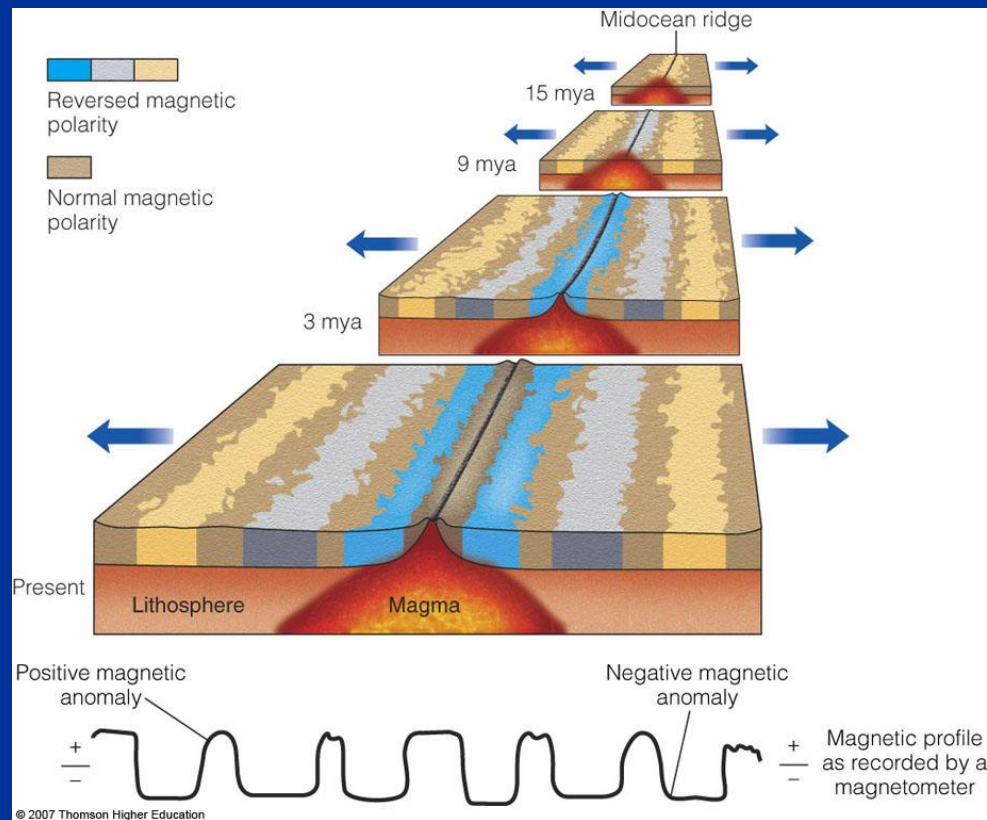
Confirmation of Hess's Hypothesis

- In addition to mapping mid-ocean ridges,
 - ocean research also revealed
 - magnetic anomalies on the sea floor
- A magnetic anomaly is a deviation
 - from the average strength
 - of Earth's Magnetic field



Confirmation of Hess's Hypothesis

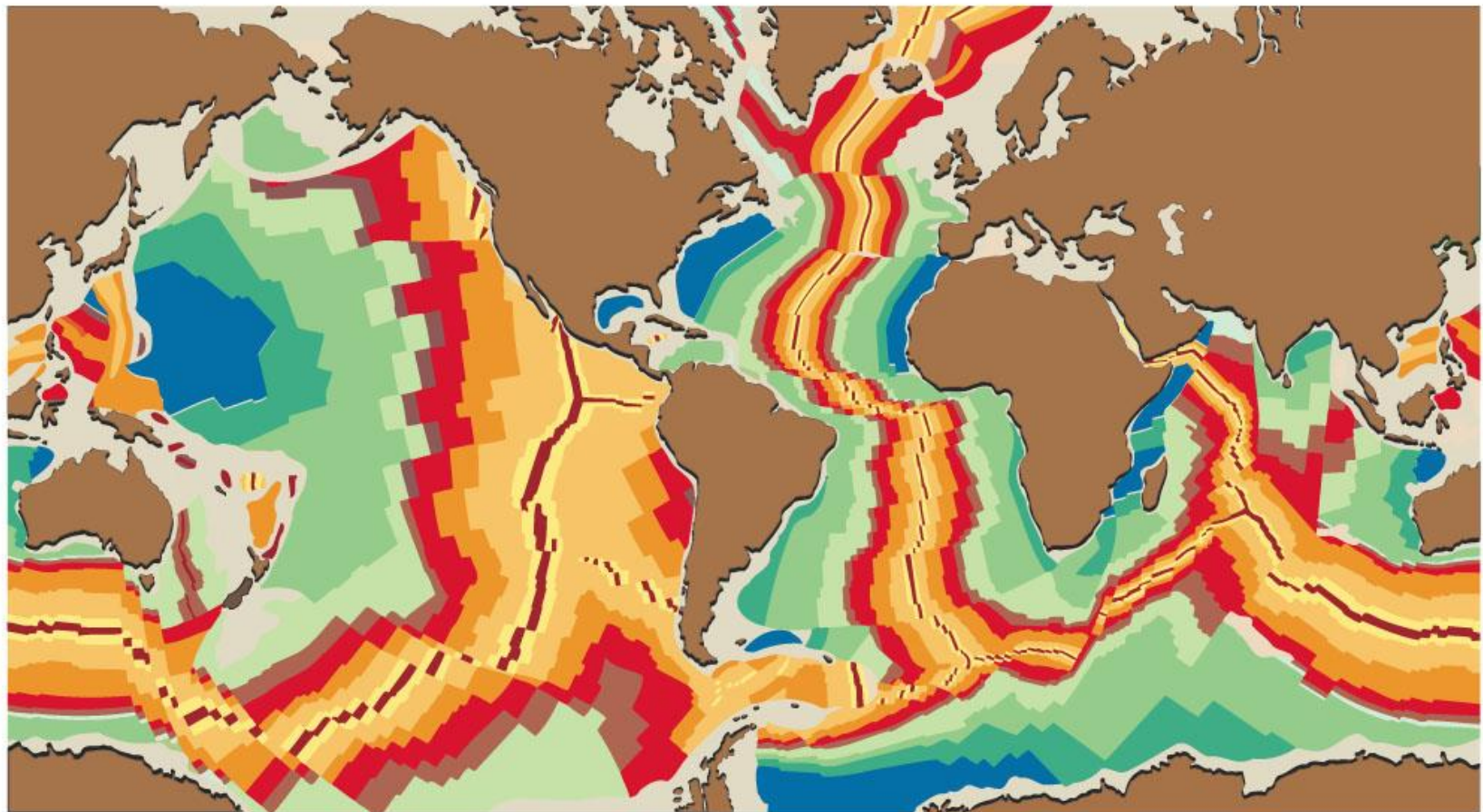
- The magnetic anomalies were discovered to be parallel and symmetrical with the oceanic ridges



Oceanic Crust Is Young

- Seafloor spreading theory indicates that
 - oceanic crust is geologically young because
 - it forms during spreading
 - and is destroyed during subduction
- Radiometric dating confirms
 - the oldest oceanic crust
 - is less than 180 million years old
- whereas oldest continental crust
 - is 3.96 billion years old

Age of Ocean Basins



Pleistocene to Recent (0–1.6 MYA)

Pliocene (1.6–5 MYA)

Miocene (5–24 MYA)

Oligocene (24–37 MYA)

Eocene (37–58 MYA)

Paleocene (58–66 MYA)

Late Cretaceous (66–88 MYA)

Middle Cretaceous (88–118 MYA)

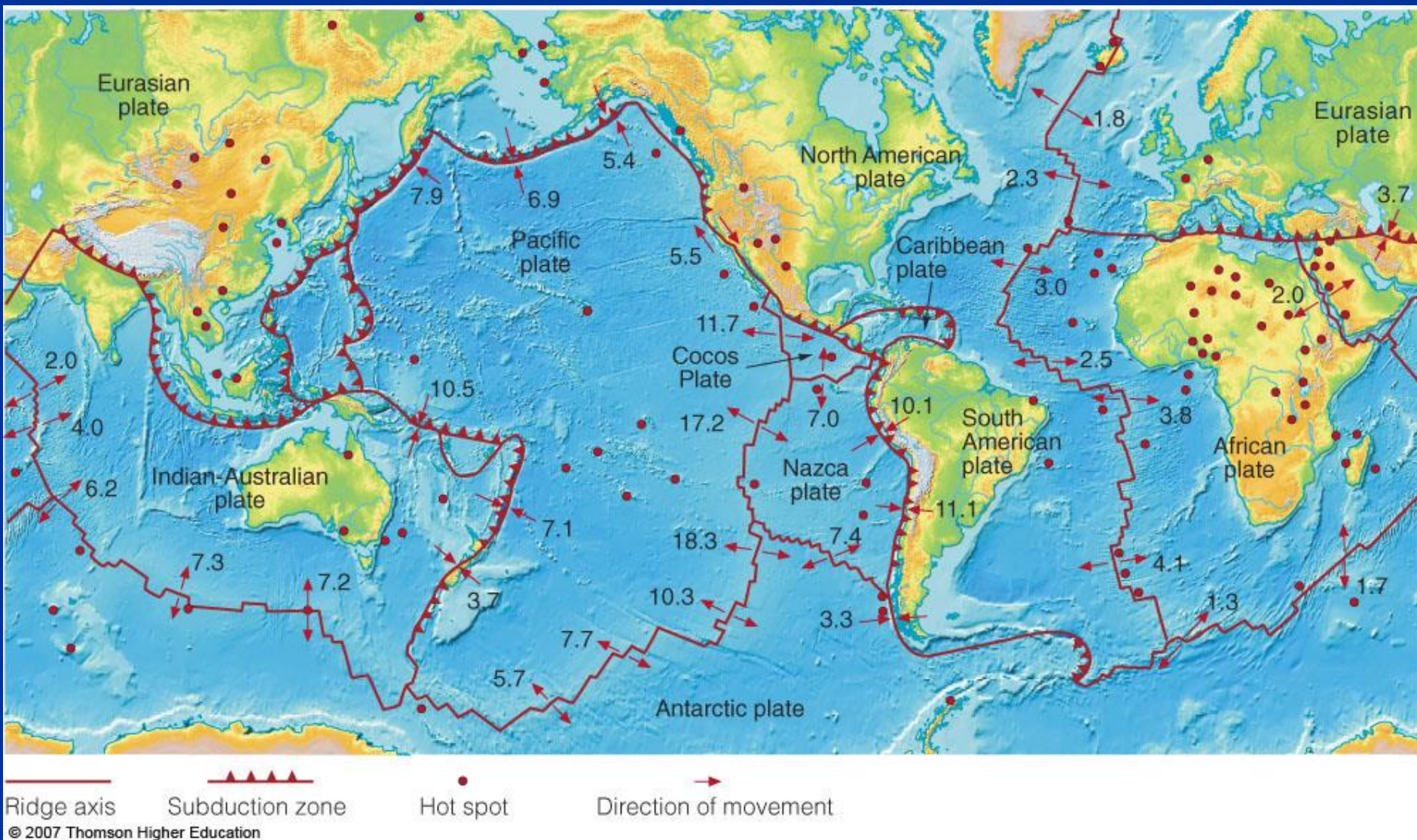
Early Cretaceous (118–144 MYA)

Late Jurassic (144–161 MYA)

Plate Tectonics

- Plate tectonic theory is based on the simple model that
 - the lithosphere is rigid
 - it consists of oceanic and continental crust with upper mantle
 - it consists of variable-sized pieces called plates
 - with plate regions containing continental crust
 - up to 250 km thick
 - and plate regions containing oceanic crust
 - up to 100 km thick

Plate Map



■ Numbers represent average rates of relative movement, cm/yr

Plate Tectonics and Boundaries

- The lithospheric plates overlie hotter and weaker semiplastic asthenosphere
- Movement of the plates
 - results from some type of heat-transfer system within the asthenosphere
- As plates move over the asthenosphere
 - they separate, mostly at oceanic ridges
 - they collide, in areas such as oceanic trenches
 - where they may be subducted back into the mantle

Divergent Boundaries

- Divergent plate boundaries
 - or *spreading ridges*, occur
 - where plates are separating
 - and new oceanic lithosphere is forming.
- Crust is extended
 - thinned and fractured
- The magma
 - originates from partial melting of the mantle
 - is basaltic
 - intrudes into vertical fractures to form dikes
 - or is extruded as lava flows

Divergent Boundaries

- Successive injections of magma
 - cool and solidify
 - form new oceanic crust
 - record the intensity and orientation
 - of Earth's magnetic field
- Divergent boundaries most commonly
 - occur along the crests of oceanic ridges
 - such as the Mid-Atlantic Ridge
- Ridges have
 - rugged topography resulting from displacement
 - of rocks along large fractures
 - shallow earthquakes

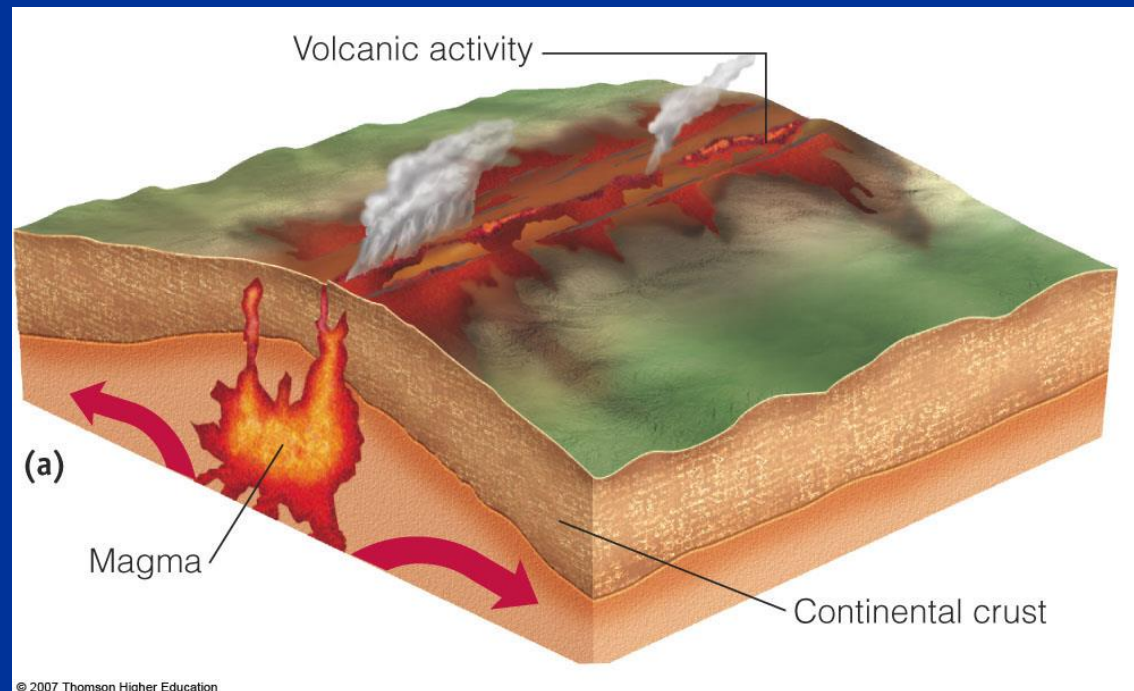
Divergent Boundaries

- Ridges also have
 - high heat flow
 - and basaltic flows or pillow lavas
- Pillow lavas have
 - a distinctive bulbous shape resulting from underwater eruptions



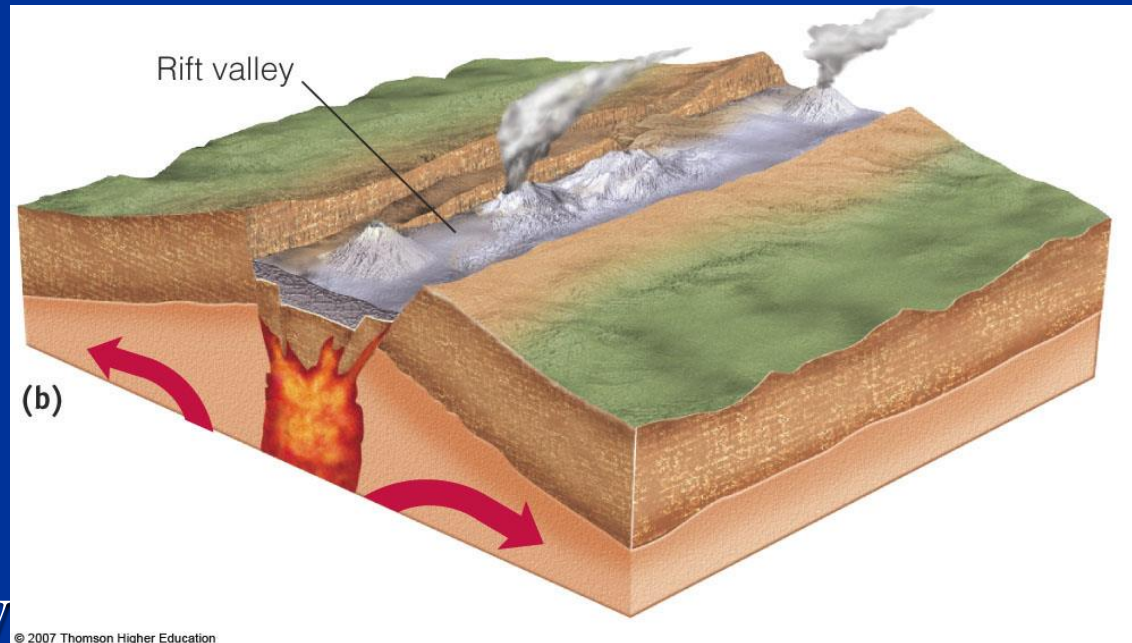
Divergent Boundaries

- Divergent boundaries are also present
 - under continents during the early stages
 - of continental breakup
- Beneath a continent,
 - magma wells up, and
 - the crust is initially
 - elevated,
 - stretched
 - and thinned



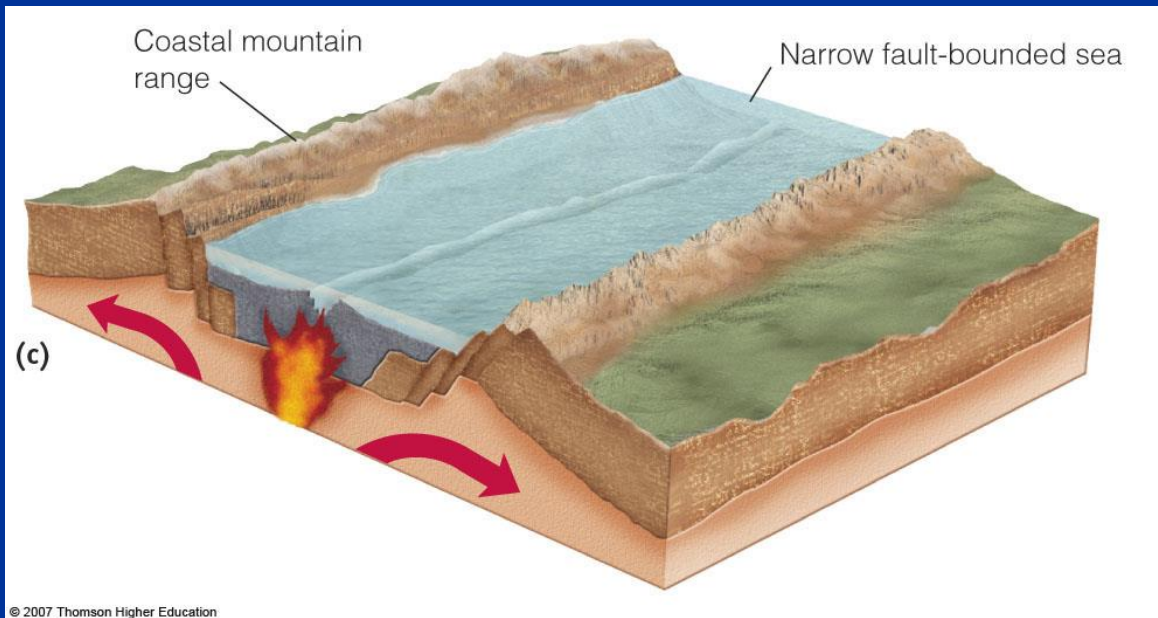
Rift Valley

- The stretching produces fractures and rift valleys.
- During this stage,
 - magma typically
 - intrudes into the fractures
 - and flows onto the valley floor
- Example: East African Rift Valley



Narrow Sea

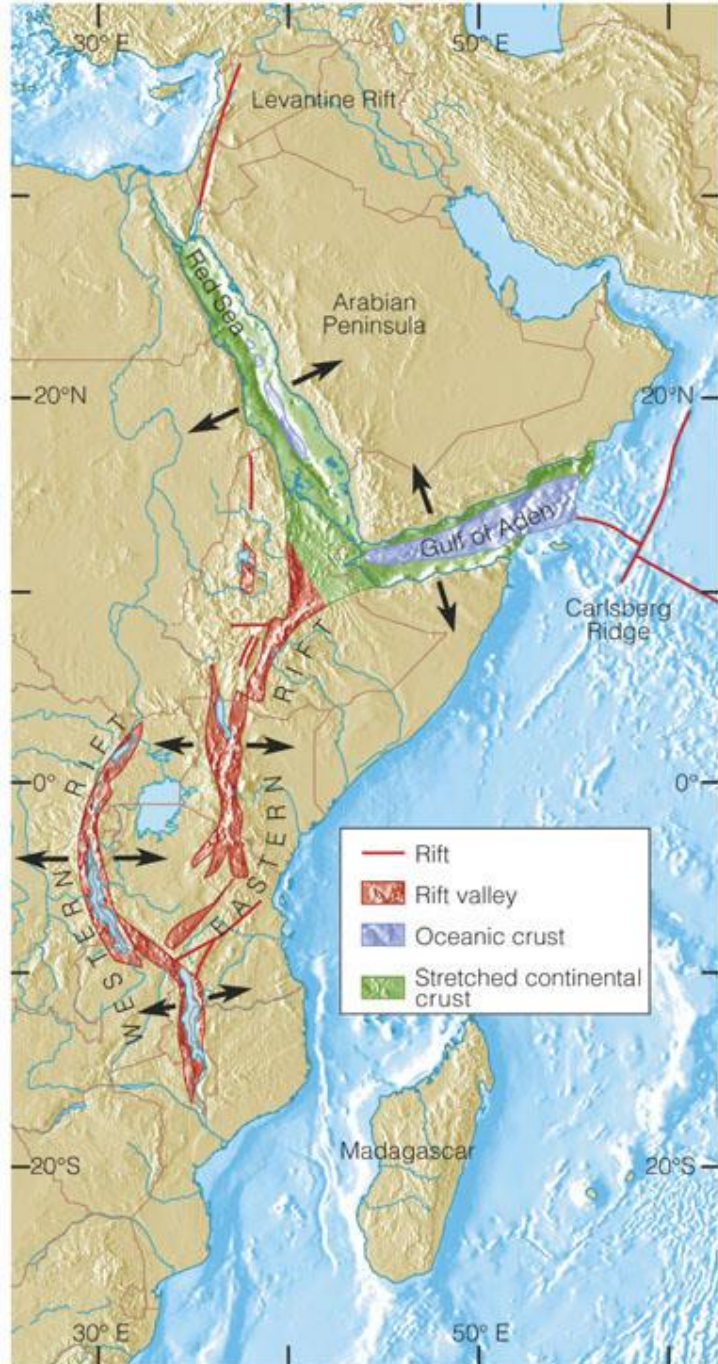
- As spreading proceeds, some rift valleys
 - will continue to lengthen and deepen until



- the continental crust eventually breaks
- a narrow linear sea is formed,
- separating two continental blocks
- Examples:
 - Red Sea
 - Gulf of California

Modern Divergence

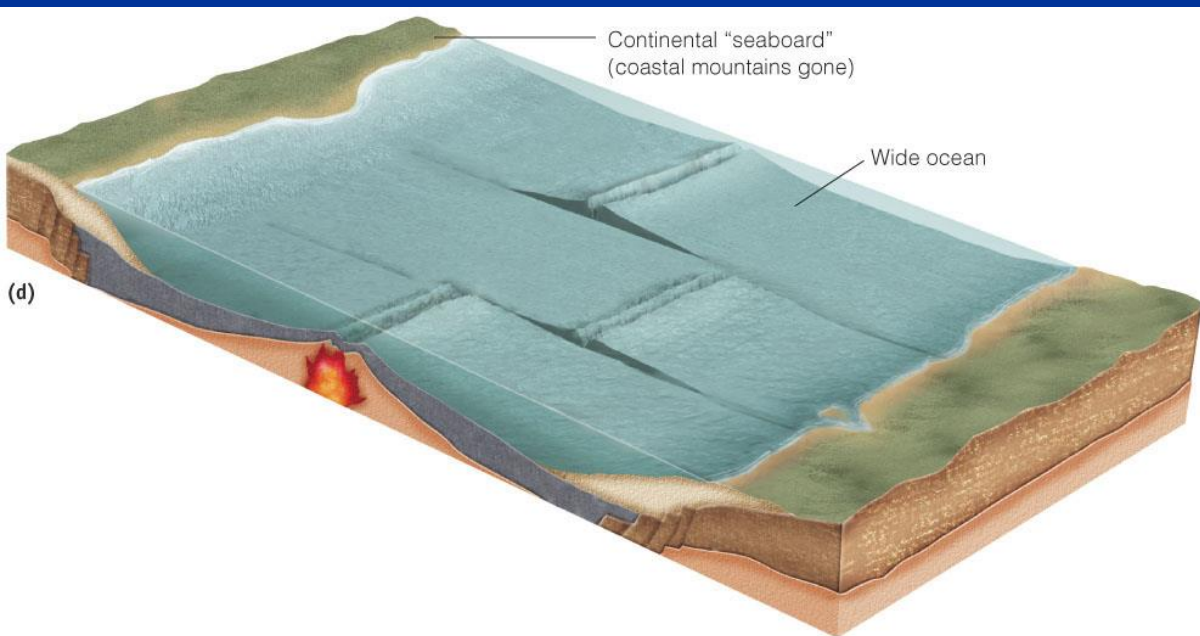
- View looking down the Great Rift Valley of Africa.



Ocean

- As a newly created narrow sea
 - continues to spread,
 - it may eventually become
 - an expansive ocean basin

- such as the Atlantic Ocean basin is today,
 - separating North and South America
 - from Europe and Africa
 - by thousands of kilometers



Atlantic Ocean Basin



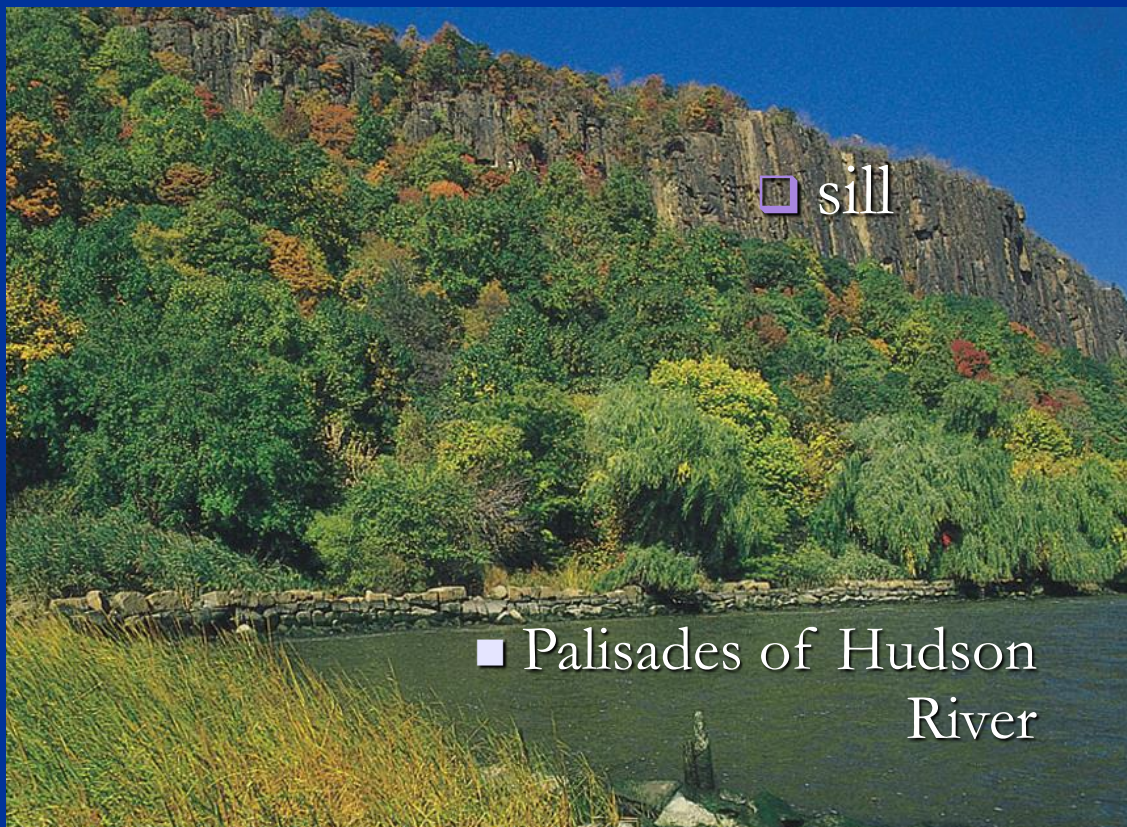
An Example of Ancient Rifting

- What features in the rock record can geologists use to recognize ancient rifting?
 - faults
 - dikes
 - sills
 - lava flows
 - thick sedimentary sequences within rift valleys
- Example:
 - Triassic fault-block basins in eastern US



Ancient Rifting

- These Triassic fault basins
 - mark the zone of rifting
 - between North America and Africa



- They contain thousands of meters of continental sediment
- and are riddled with dikes and sills

Convergent Boundaries

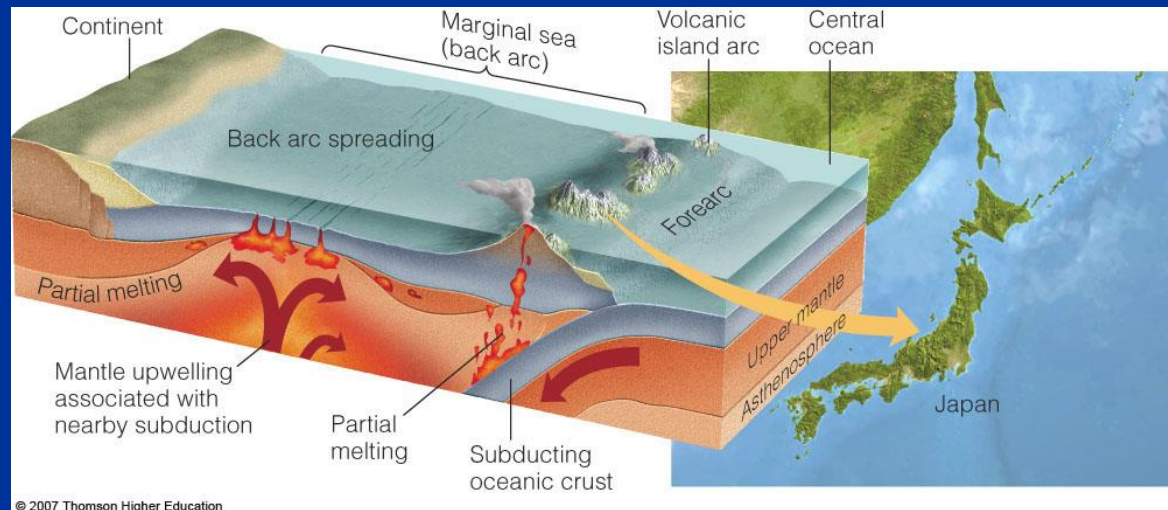
- Older crust must be destroyed
 - at convergent boundaries
 - so that Earth's surface area remains the same
- Where two plates collide,
 - subduction occurs
 - when an oceanic plate
 - descends beneath the margin of another plate
 - The subducting plate
 - moves into the asthenosphere
 - is heated
 - and eventually incorporated into the mantle

Convergent Boundaries

- Convergent boundaries are characterized by
 - deformation
 - volcanism
 - mountain building
 - metamorphism
 - earthquake activity
 - valuable mineral deposits
- Convergent boundaries are of three types:
 - oceanic-oceanic
 - oceanic-continental
 - continental-continental

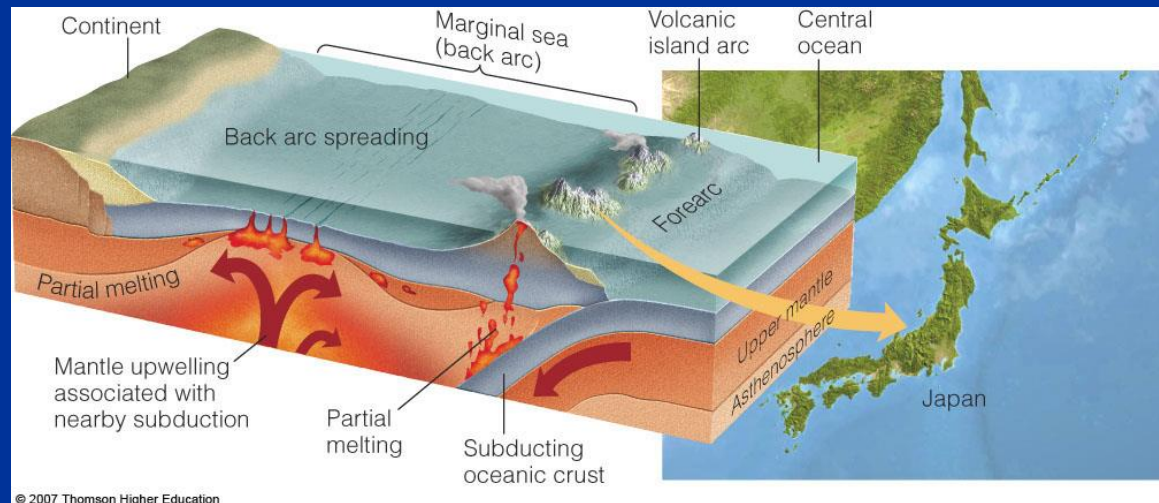
Oceanic-Oceanic Boundary

- When two oceanic plates converge,
 - one is subducted beneath the other
 - along an oceanic-oceanic plate boundary
 - forming an oceanic trench
 - and a *subduction complex*
 - composed of slices of folded and faulted sediments
 - and oceanic lithosphere
 - scraped off the descending plate



Volcanic Island Arc

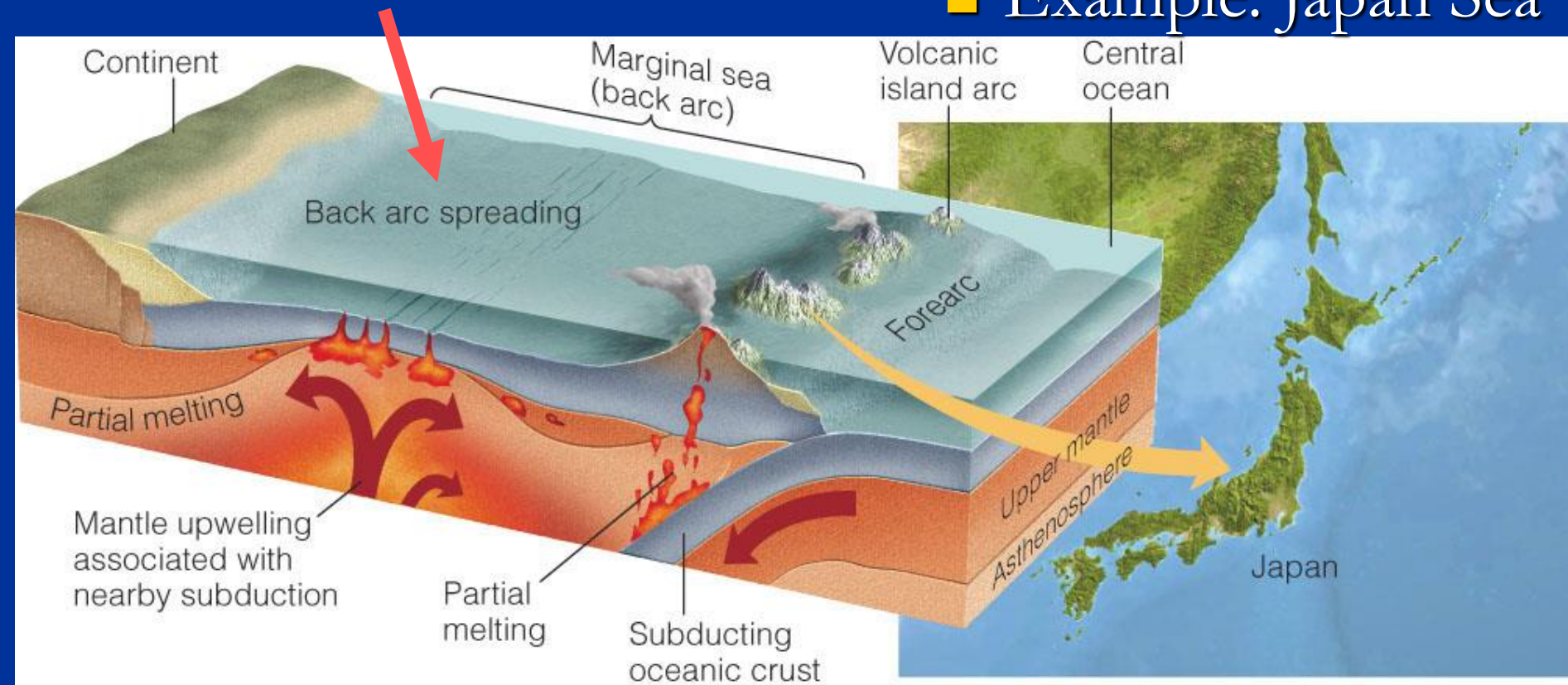
- As the plate subducts into the mantle,
 - it is heated and partially melted
 - generating magma of ~ andesitic composition
 - that rises to the surface
 - because it is less dense than the surrounding mantle rocks
- At the surface of the non-subducting plate,
 - the magma forms a volcanic island arc



Oceanic-Oceanic Plate Boundary

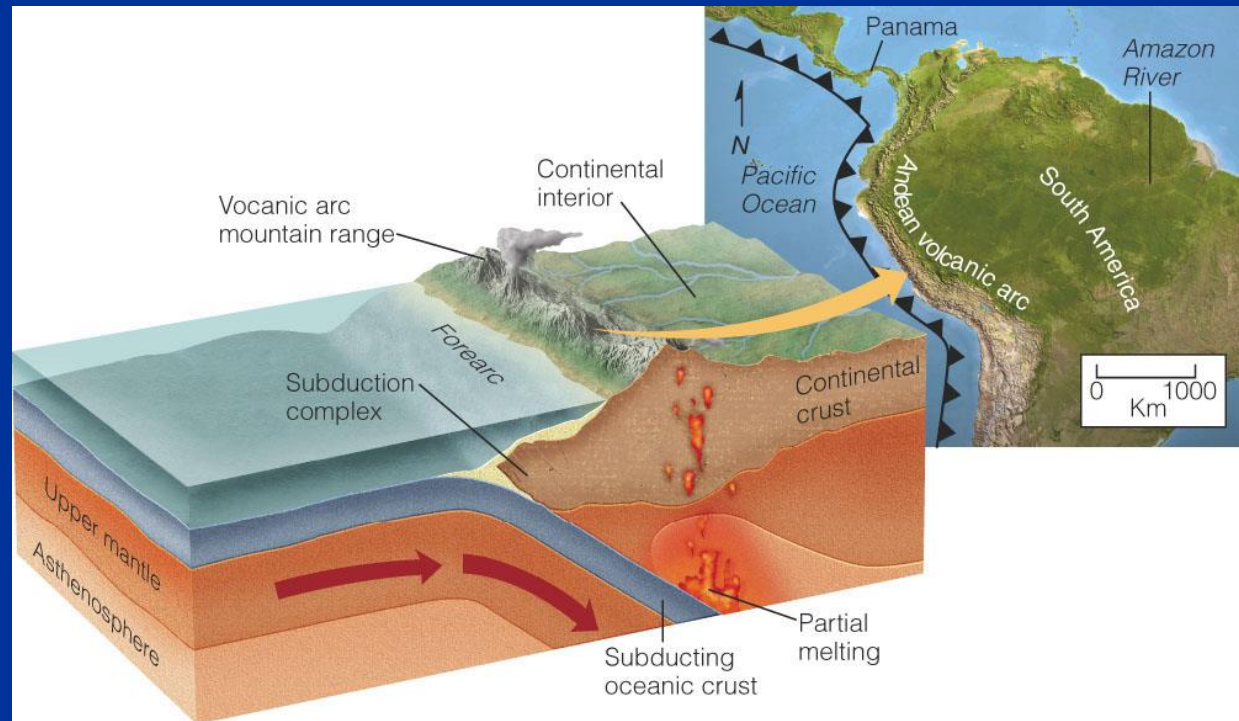
- A *back-arc basin* forms in some cases of fast subduction.
 - The lithosphere on the landward side of the island arc
 - is stretched and thinned

■ Example: Japan Sea



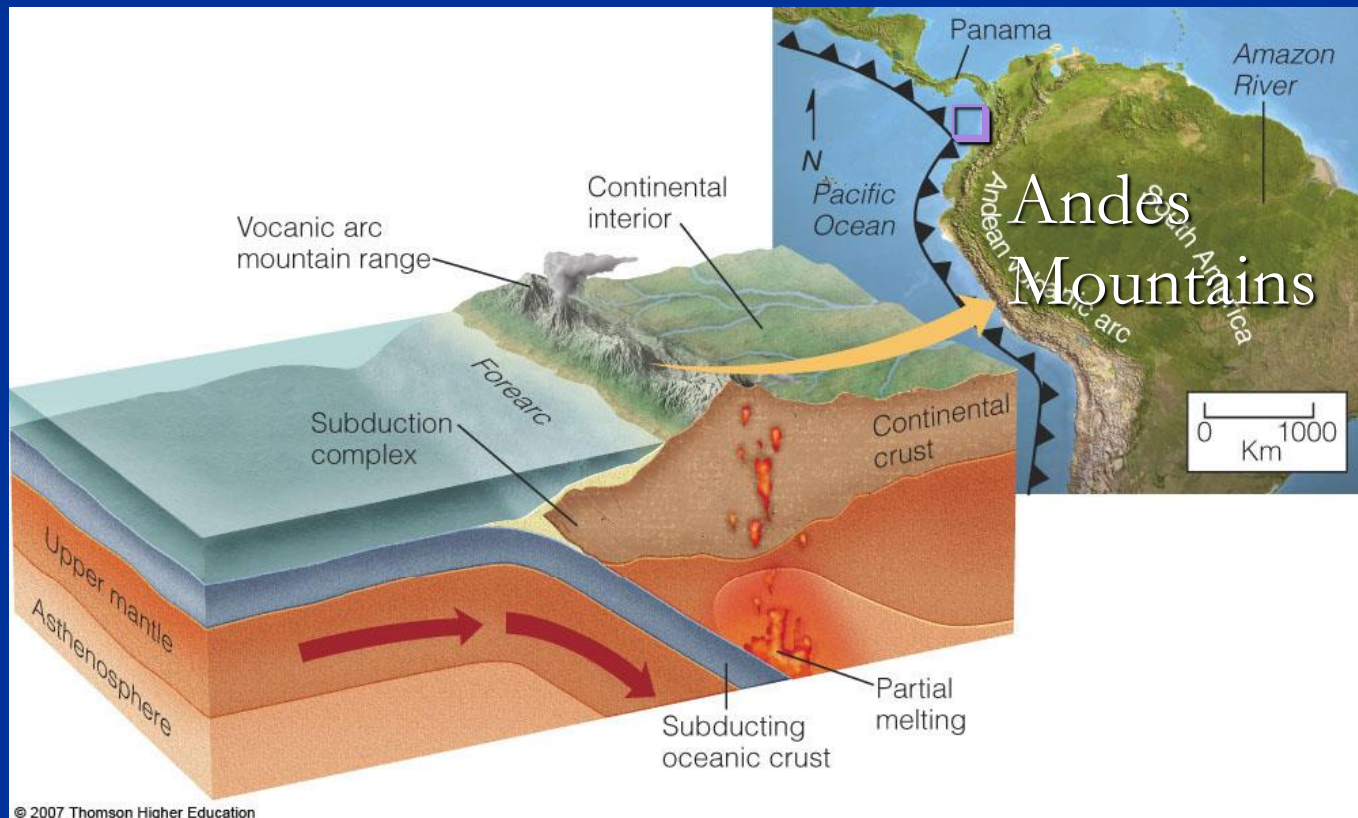
Oceanic-Continental Boundary

- An oceanic-continental plate boundary
 - occurs when a denser oceanic plate
 - subducts under less dense continental lithosphere
- Magma generated by subduction
 - rises into the continental crust to form large igneous bodies
 - or erupts to form a volcanic arc of andesitic volcanoes
 - Example: Pacific coast of South America



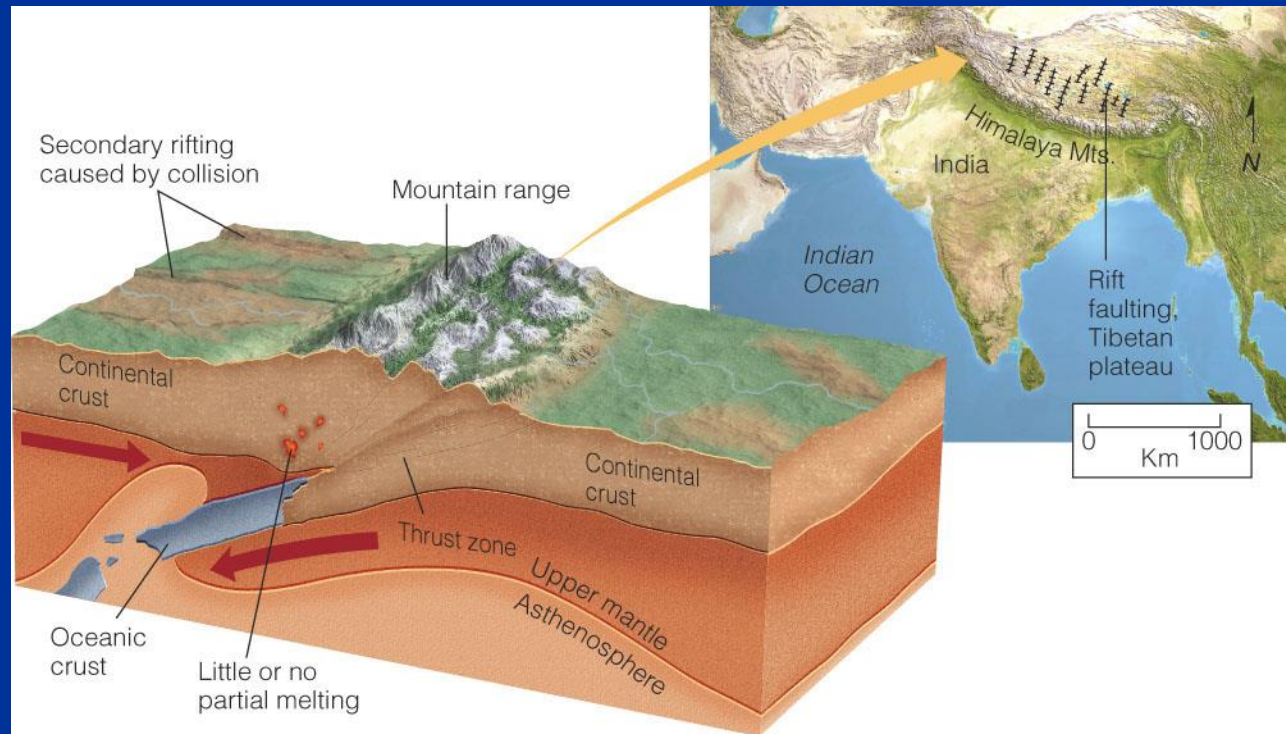
Oceanic-Continental Boundary

- Where the Nazca plate in the Pacific Ocean is subducting under South America
 - the Peru-Chile Trench marks subduction site
 - and the Andes Mountains are the volcanic arc



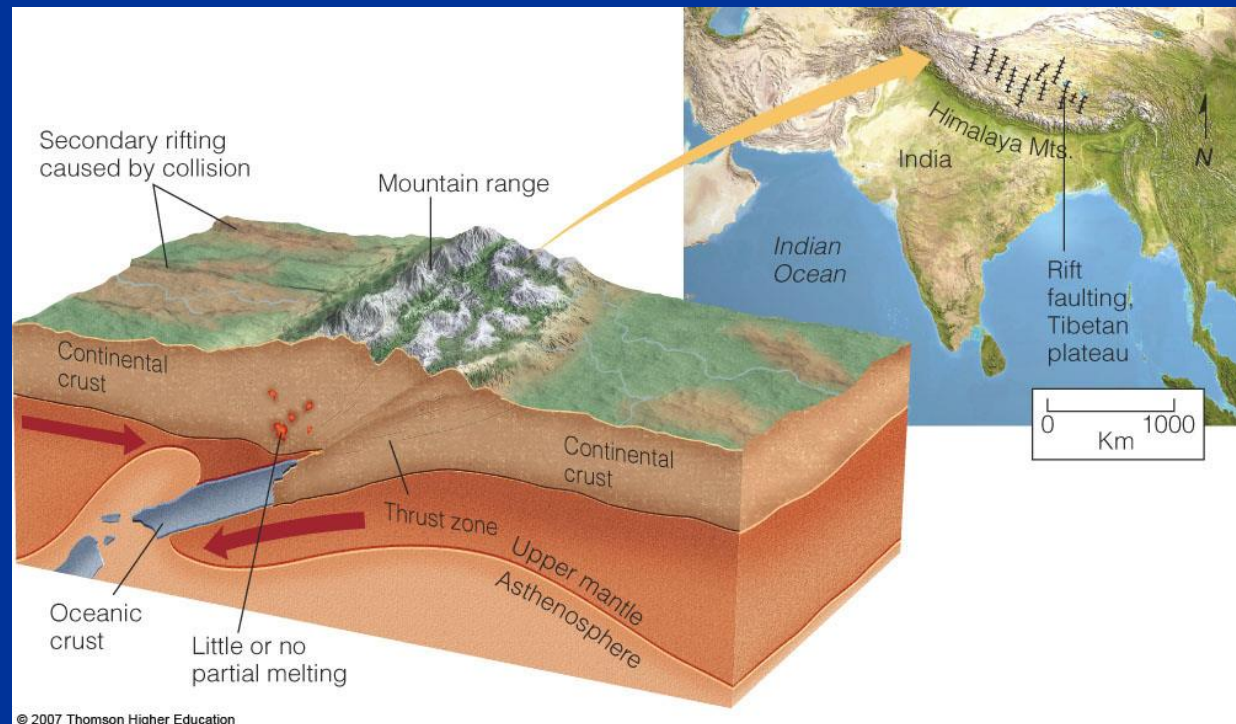
Continent-Continent Boundary

- Two approaching continents are initially
 - separated by ocean floor that is being subducted
 - under one of them, which, thus, has a volcanic arc
- When the 2 continents collide
 - the continental lithosphere cannot subduct
- Its density is too low,
 - although one continent may partly slide under the other



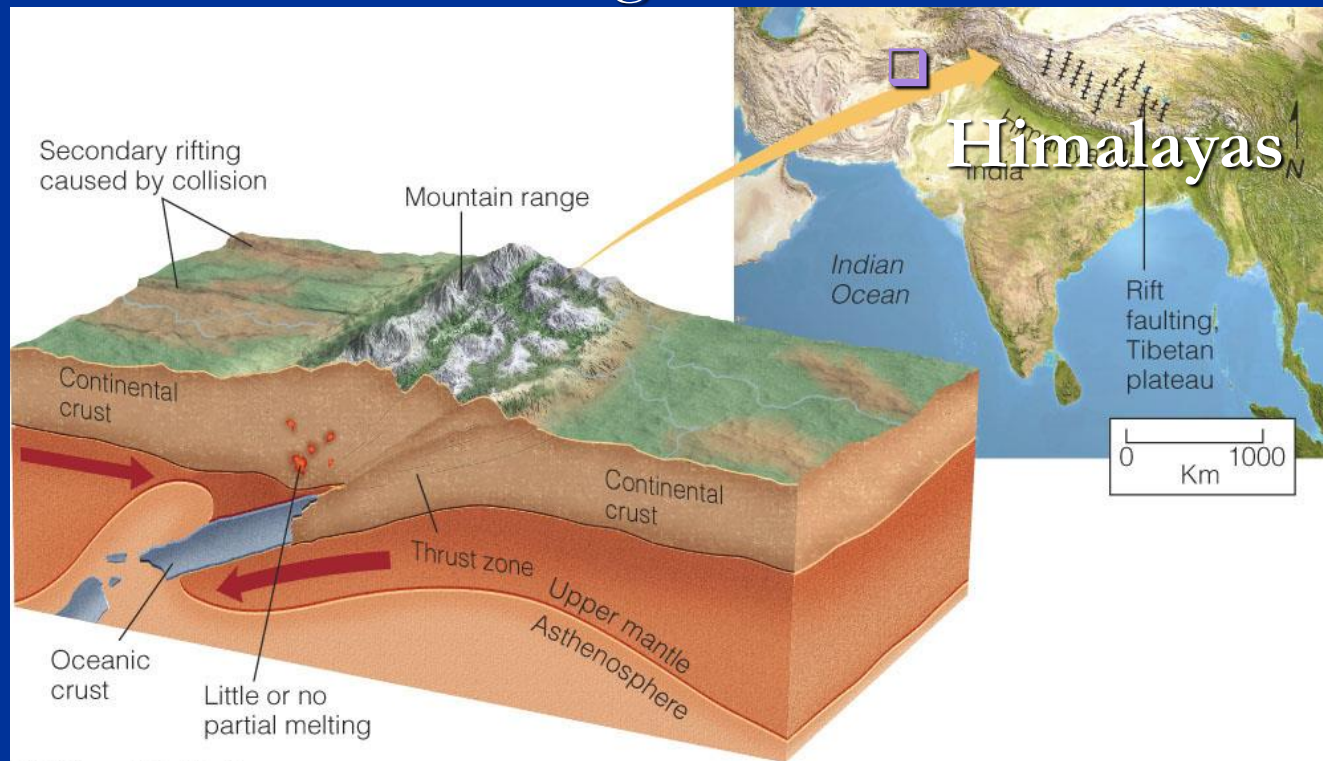
Continent-Continent Boundary

- When the 2 continents collide
 - they weld together at a continent-continent plate boundary,
 - where an interior mountain belt forms consisting of
 - deformed sedimentary rocks
 - igneous intrusions
 - metamorphic rocks
 - fragments of oceanic crust
- Earthquakes occur here



Continental-Continental Boundary

- Example: Himalayas in central Asia
 - Earth's youngest and highest mountain system
 - resulted from collision between India and Asia
 - began 40 to 50 million years ago
 - and is still continuing

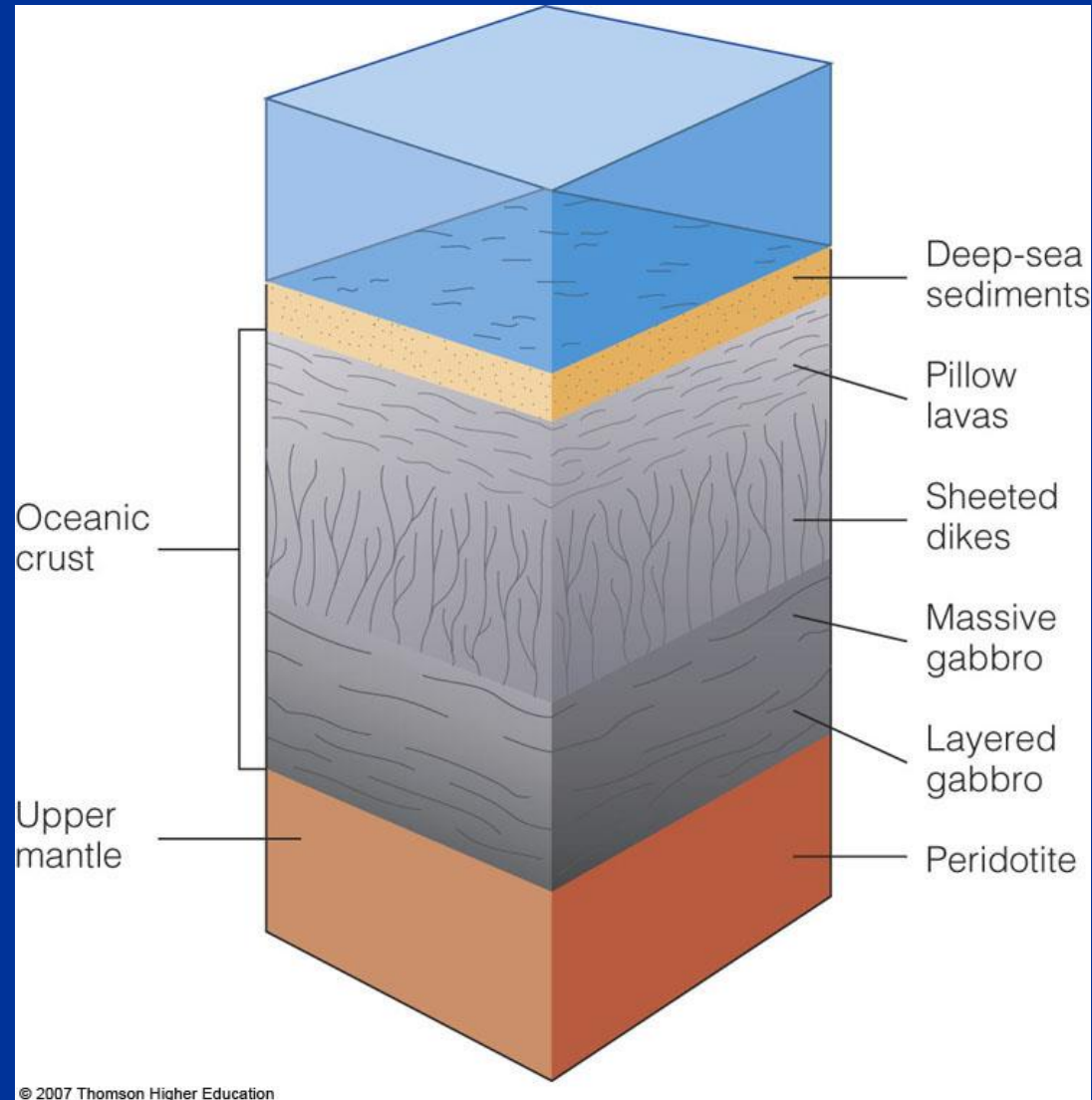


Recognizing Ancient Convergent Boundaries

- How can former subduction zones be recognized in the rock record?
 - Andesitic magma erupted,
 - forming island arc volcanoes and continental volcanoes
 - The subduction complex results in
 - a zone of intensely deformed rocks
 - between the trench and the area of igneous activity
 - Sediments and submarine rocks
 - are folded, faulted and metamorphosed
 - making a chaotic mixture of rocks termed a *mélange*
 - Slices of oceanic lithosphere may be accreted
 - to the continent edge and are called ophiolites

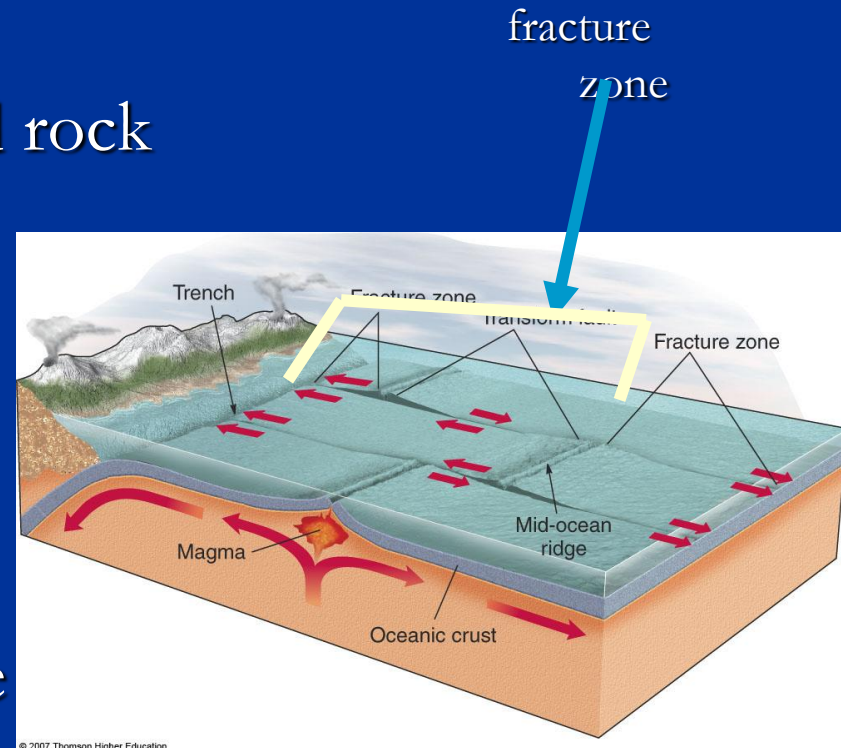
Ophiolite

- Ophiolites consist of layers
 - ▣ representing parts of the oceanic crust and upper mantle.
- The sediments include
 - ▣ graywackes
 - ▣ black shales
 - ▣ cherts
- Ophiolites are key to detecting old subduction zones



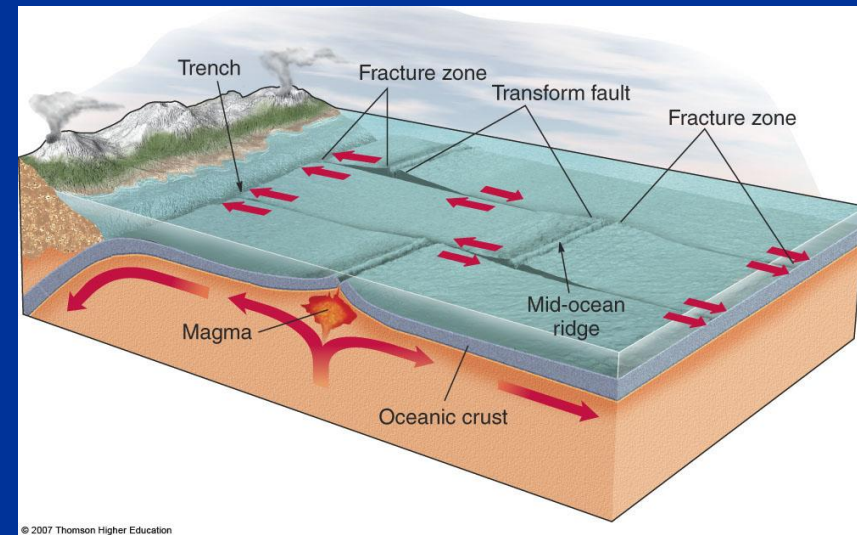
Transform Boundaries

- The third type of plate boundary is a transform plate boundary
 - where plates slide laterally past each other
 - roughly parallel to the direction of plate movement
- Movement results in
 - zone of intensely shattered rock
 - numerous shallow earthquakes
- The majority of transform faults
 - connect two oceanic ridge segments
 - and are marked by fracture zones



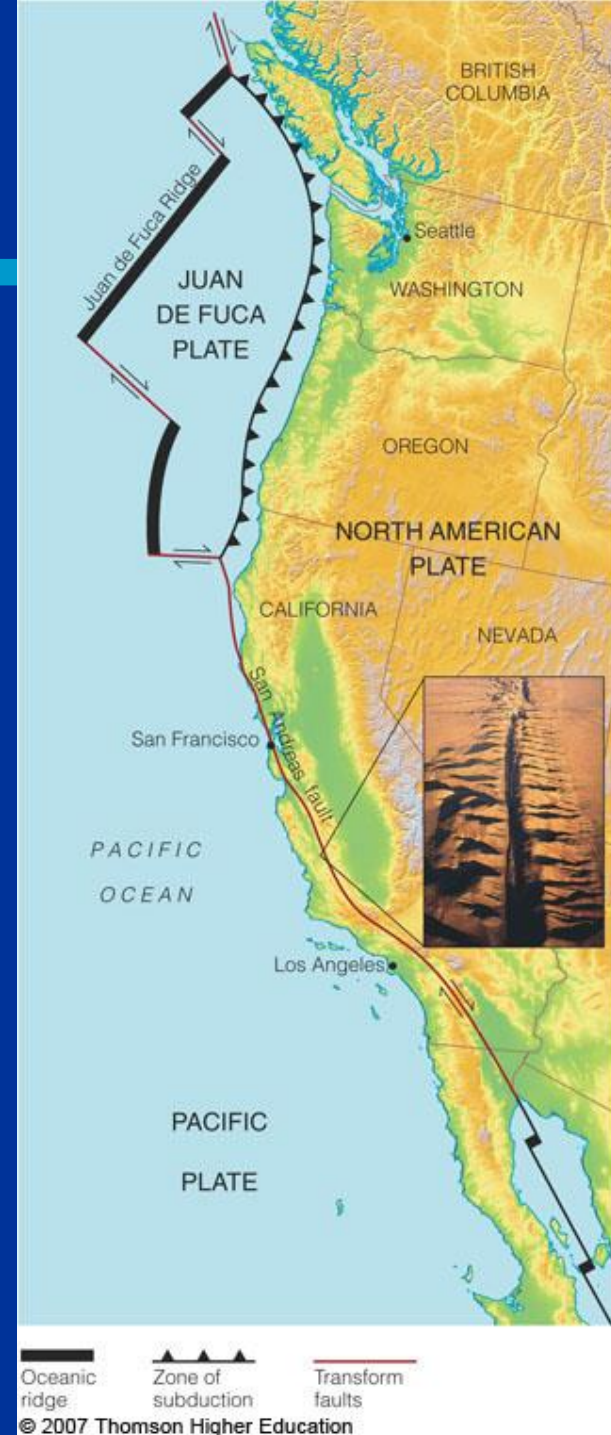
Transform Boundaries

- Other kinds of transform plate boundaries
 - connect two trenches
 - or connect a ridge to a trench
 - or even a ridge or trench to another transform fault
- Transforms can also extend into continents



Transform Boundaries

- Example: San Andreas Fault, California
 - separates the Pacific plate from the North American plate
 - connects ridges in
 - Gulf of California
 - with the Juan de Fuca and Pacific plates
 - Many of the earthquakes in California result from movement along this fault

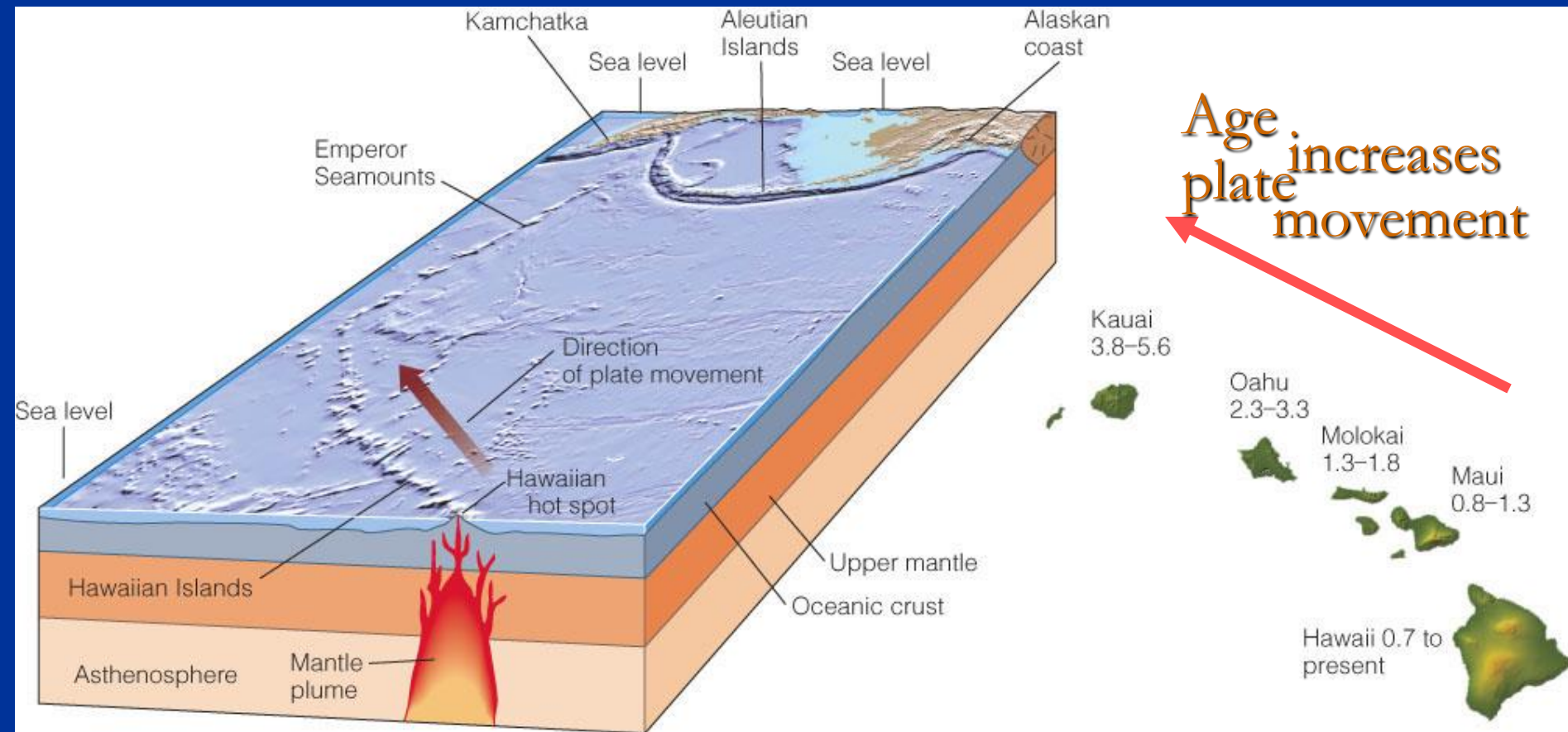


Hot Spots and Mantle Plumes

- Hot spots are locations where
 - stationary columns of magma
 - originating deep within the mantle,
 - called mantle plumes
 - slowly rise to the surface
- Mantle plumes remain stationary
 - although some evidence suggests they may move
- When plates move over them
 - hot spots leave trails
 - of extinct, progressively older volcanoes
 - called *aseismic ridges*
 - which record the movement of the plates

Hot Spots and Mantle Plumes

- Example: Emperor Seamount-Hawaiian Island chain



How Is Plate Motion Determined?

- Rates of plate movement can be calculated in several ways
 - Sediment
 - determine the age of sediment that is
 - immediately above any portion of oceanic crust
 - divide the distance from the spreading ridge by the age
 - gives average rate of movement relative to the ridge
 - LEAST ACCURATE METHOD

Plate Movement Measurements

- Seafloor magnetic anomalies
 - measure the distance of the magnetic anomaly in seafloor crust from the spreading ridge
 - divide by the age of the anomaly
- The present average rate of movement, relative motion, and the average rate of motion in the past can be determined.

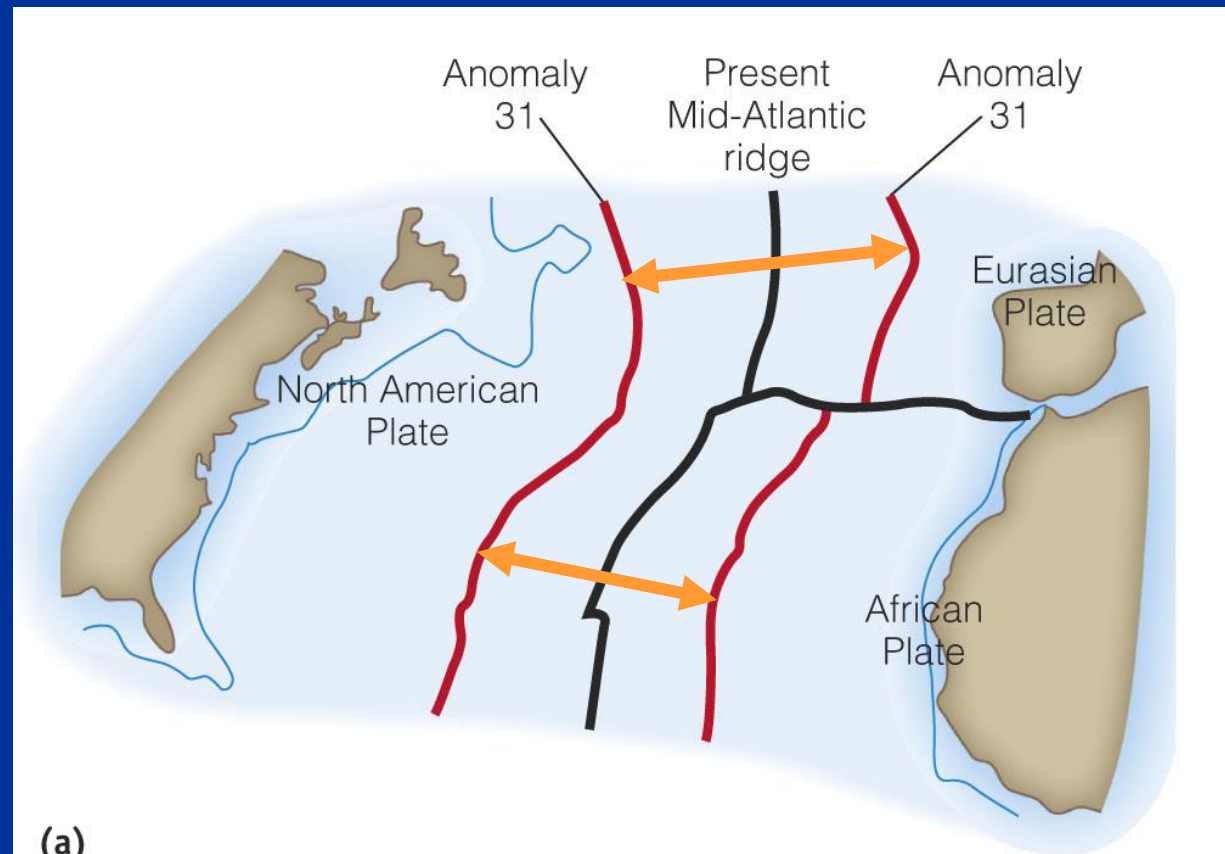
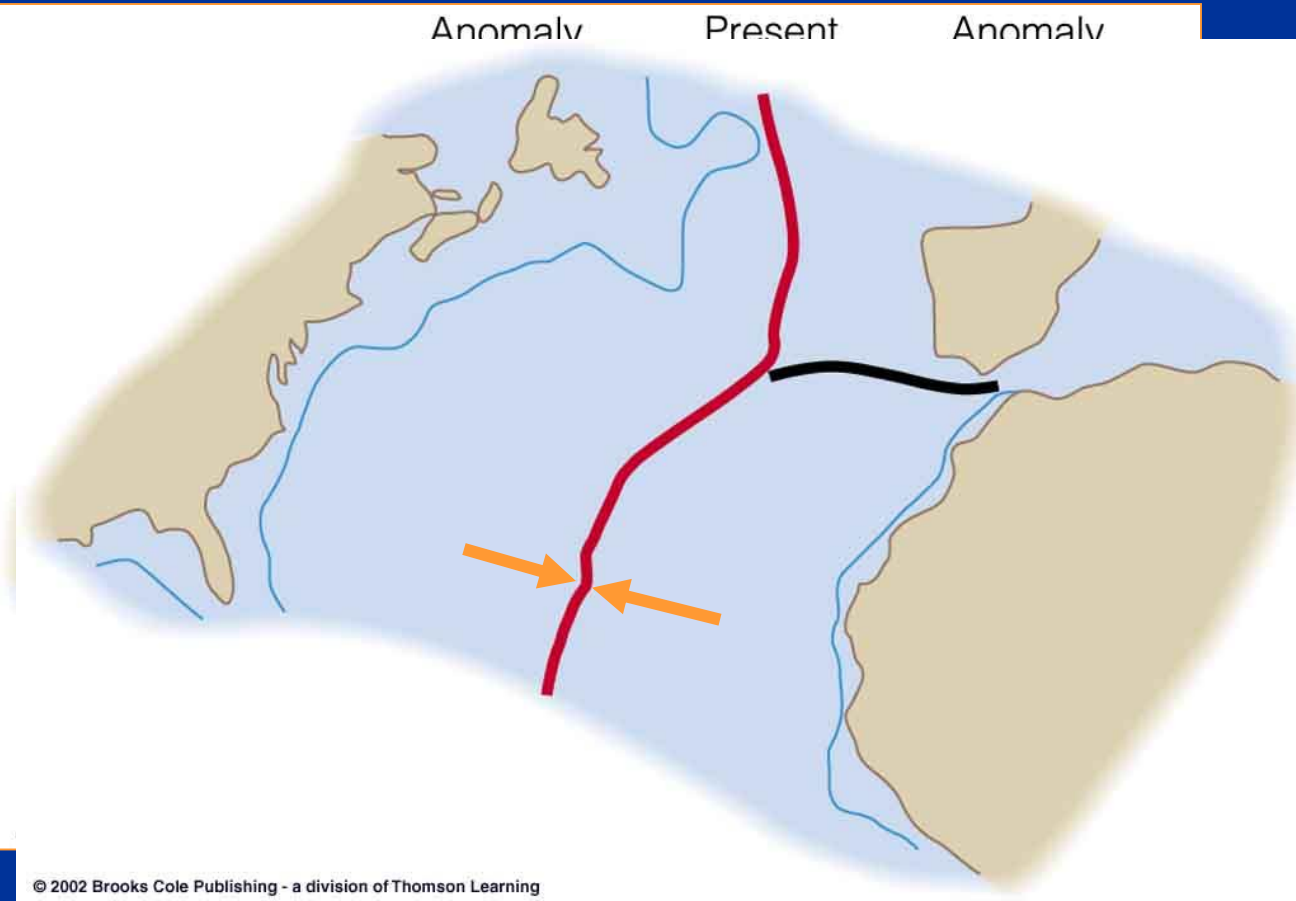


Plate Position Reconstruction

- Reconstructing plate positions
 - to determine the plate and continent positions at the time of an anomaly
 - move the anomaly back to the spreading ridge



- Since subduction destroys oceanic crust
- this kind of reconstruction cannot be done earlier than the oldest oceanic crust

Plate Movement Measurements

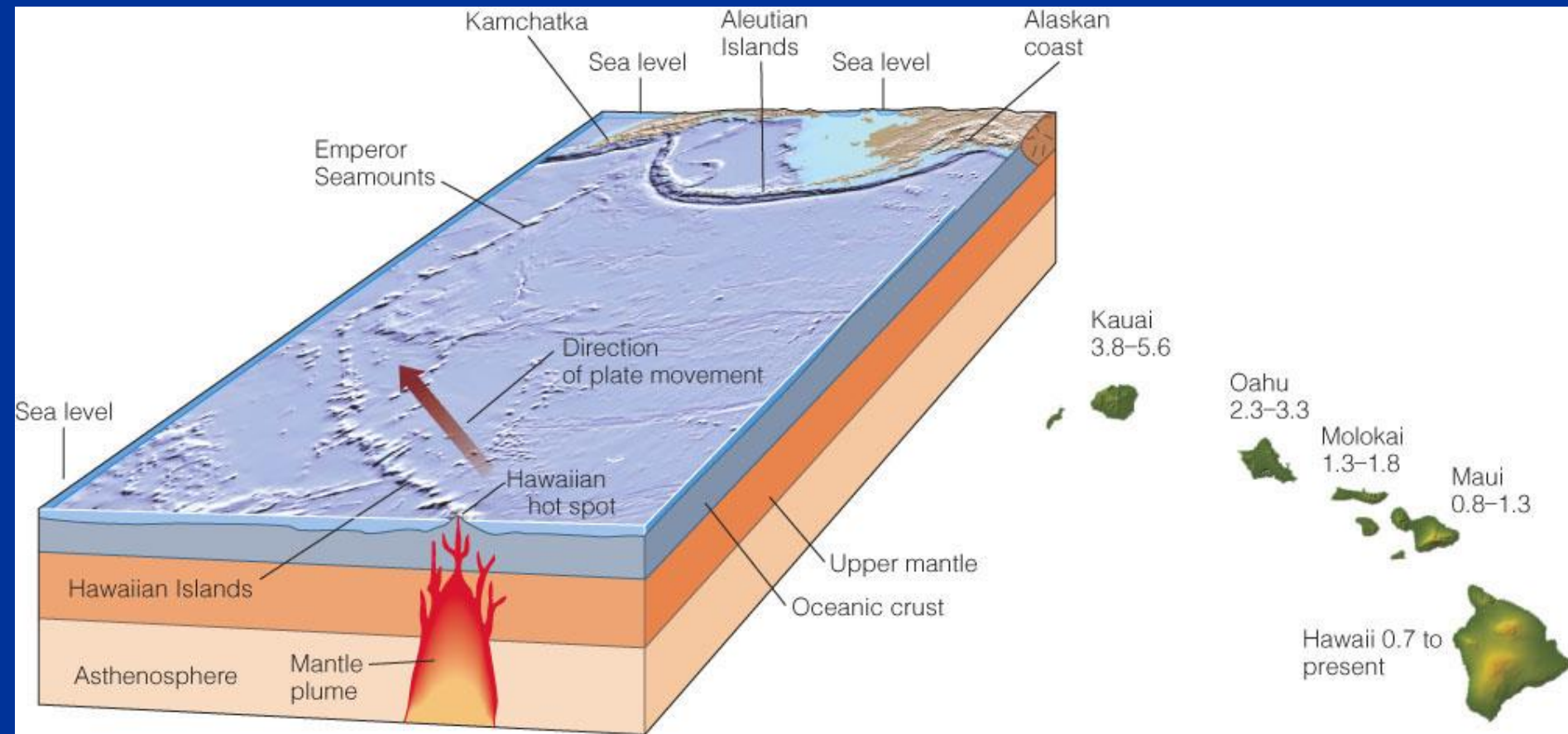
■ Satellite-laser ranging

- bounce laser beams from a station on one plate
- off a satellite, to a station on another plate
- measure the elapsed time
- after sufficient time has passed to detect motion
- measure the elapsed time again
- use the difference in elapsed times to calculate
- the rate of movement between the two plates

■ Hot spots

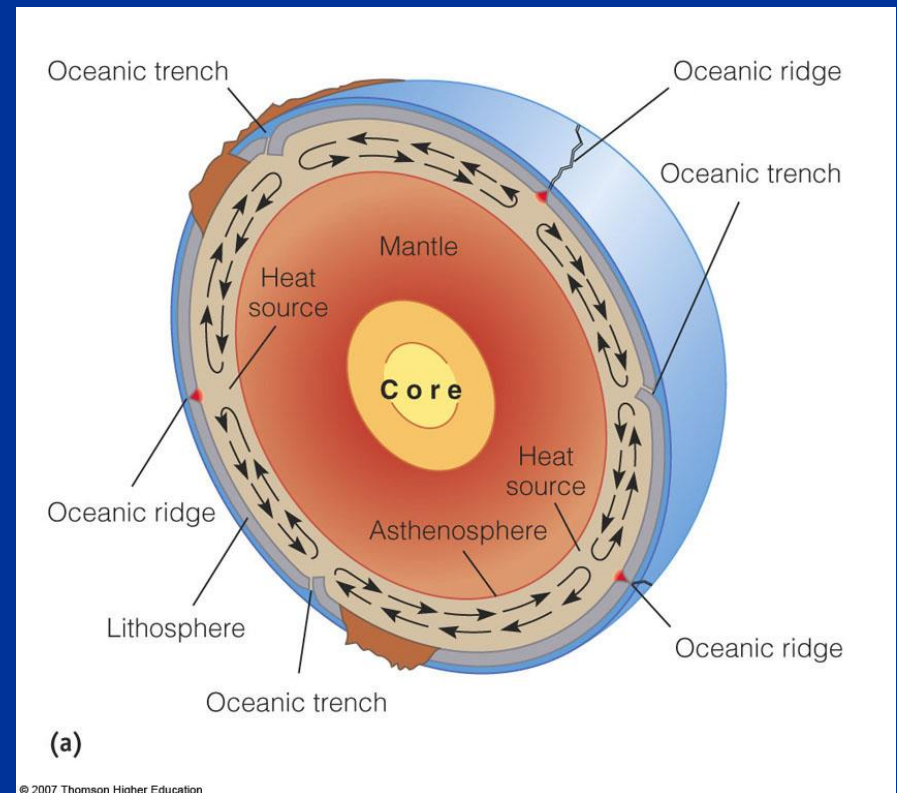
- determine the age of rocks and their distance from a hot spot
- divide the distance by the age
- this gives the motion relative to the hot spot so
- (possibly) the absolute motion of the plate

Plate Movement at Hot Spot



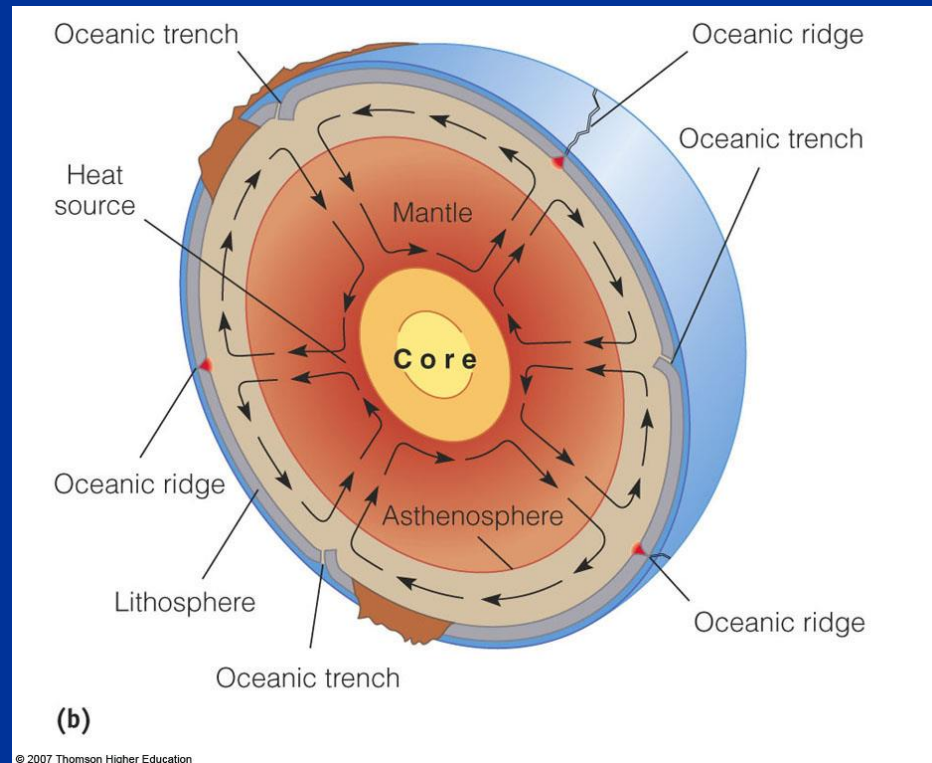
What Is the Driving Mechanism of Plate Tectonics?

- Most geologists accept some type of convective heat system
 - as the basic cause
 - of plate motion
- In one possible model,
 - thermal convection cells
 - are restricted to the asthenosphere



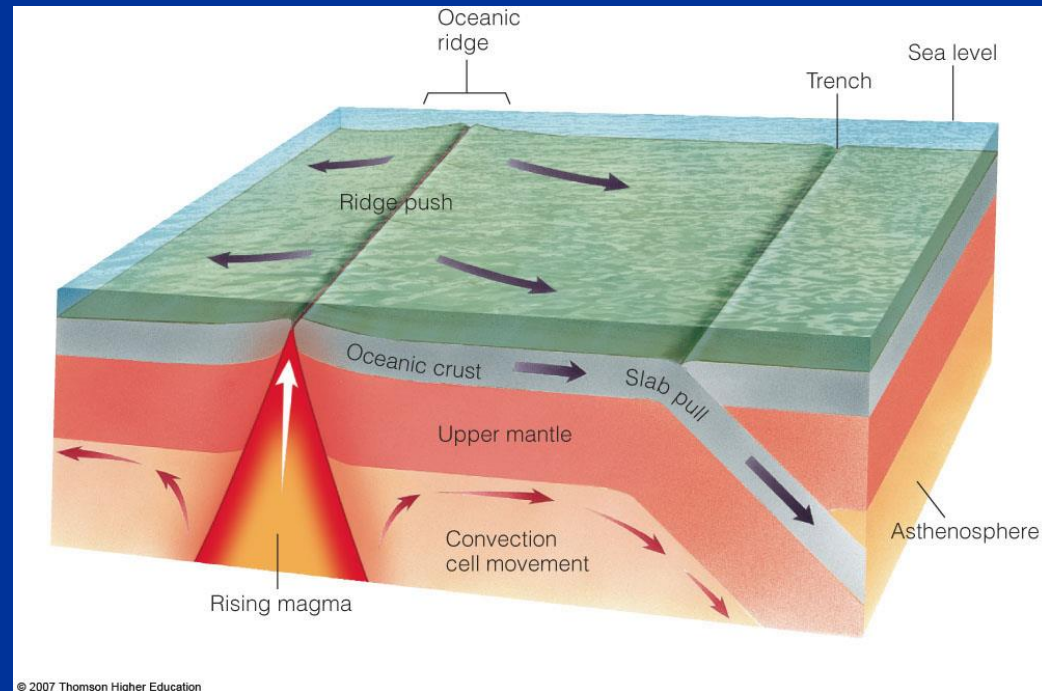
What Is the Driving Mechanism of Plate Tectonics?

- In a second model, the entire mantle is involved in thermal convection.
- In both models,
 - spreading ridges mark the rising limbs of neighboring convection cells
 - trenches occur where the convection cells descend back into Earth's interior



What Is the Driving Mechanism of Plate Tectonics?

- In addition to a thermal convection system,
 - some geologists think that movement may be aided by
 - “slab-pull”
 - the slab is cold and dense and pulls the plate
 - “ridge-push”
 - rising magma pushes the ridges up
 - and gravity pushes the oceanic lithosphere away from the ridge and toward the trench



How Are Plate Tectonics and Mountain Building Related?

- An orogeny is an episode
 - of intense rock deformation or mountain building
- It results from compressive forces
 - related to plate movement
- During subduction,
 - sedimentary and volcanic rocks
 - are folded and faulted along the plate margin
- Most orogenies occur along oceanic-continental
 - or continental-continental plate boundaries

How Are Plate Tectonics and Mountain Building Related?

- Ophiolites are evidence of ancient convergent plate boundaries
- The *Wilson Cycle* describes the relationship between mountain building and the opening and closing of ocean basins.

Terrane Tectonics

- Terranes differ from neighboring regions
 - in their fossil content,
 - stratigraphy, structural trends,
 - and paleomagnetism
- They probably formed elsewhere
 - were carried great distances as parts of other plates
 - until they collided with other terranes or continents
- Numerous terranes have been identified in mountains of the North American Pacific coast region

How Does Plate Tectonics Affect the Distribution of Life?

- Present distribution of plants and animals
 - is largely controlled by climate
 - and geographic barriers
- Barriers create *biotic provinces*
 - each province is a region characterized
 - by a distinctive assemblage of plants and animals
- Plate movements largely control barriers
 - When continents break up, new provinces form
 - When continents come together, fewer provinces result
 - As continents move north or south they move across temperature barriers

How Does Plate Tectonics Affect the Distribution of Life?

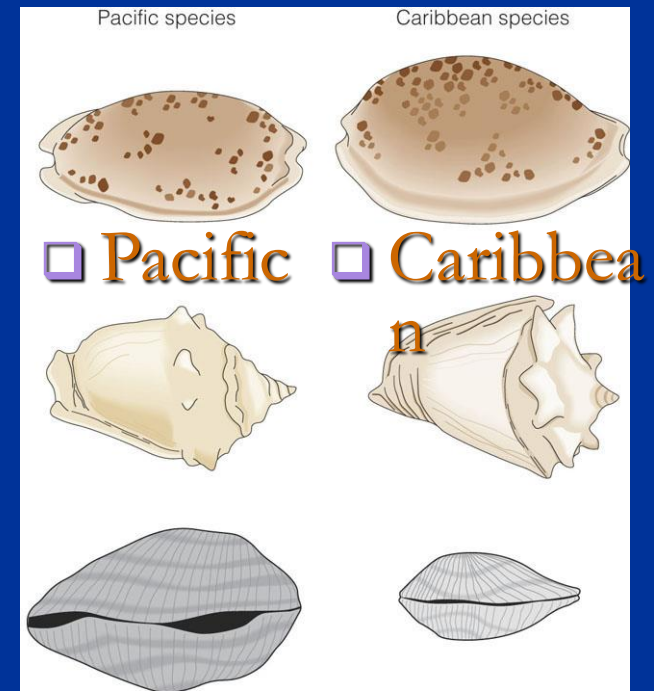
- Physical barriers caused by plate movements include
 - intraplate volcanoes
 - island arcs
 - mid-ocean ridges
 - mountain ranges
 - subduction zones

- Example: Isthmus of Panama creates a barrier to marine organisms



(a)

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(b)

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Plate Tectonics and the Distribution of Natural Resources

- Plate movements influence the formation and distribution of some natural resources such as
 - petroleum
 - natural gas
 - some mineral deposits
- Metal resources related to igneous and associated hydrothermal activity include
 - copper
 - gold
 - lead
 - silver
 - tin
 - zinc

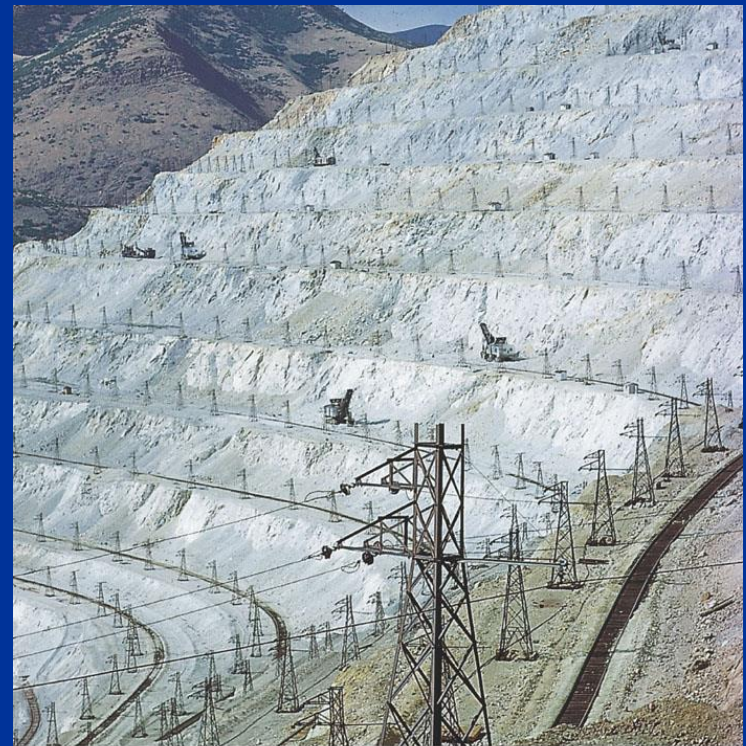
Plate Tectonics and the Distribution of Natural Resources

- Magma generated by subduction can precipitate and concentrate metallic ores
 - Example: copper deposits in western Americas
 - Bingham Mine in Utah is a huge open-pit copper mine



(a)

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(b)

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Plate Tectonics and the Distribution of Natural Resources

- Another place where hydrothermal activity
 - can generate rich metal deposits
 - is divergent boundaries
- Example: island of Cyprus in the Mediterranean
 - Copper concentrations there formed as a result
 - of precipitation adjacent to hydrothermal vents
 - along a divergent plate boundary
- Example: Red Sea
 - copper, gold, iron, lead, silver ,and zinc deposits
 - are currently forming as sulfides in the Red Sea,
 - a divergent boundary